



US 20250286801A1

(19) **United States**

(12) **Patent Application Publication**
Love et al.

(10) **Pub. No.: US 2025/0286801 A1**
(43) **Pub. Date: Sep. 11, 2025**

(54) **DECENTRALIZED COMPUTING NETWORKS, ARCHITECTURES AND TECHNIQUES FOR PROCESSING EVENTS ACROSS MULTIPLE CHANNELS**

H04L 12/14 (2024.01)
H04L 41/0896 (2022.01)
H04L 41/0897 (2022.01)
H04L 47/12 (2022.01)

(71) Applicant: **SURGETECH M LLC**, Lago (PT)

(52) **U.S. Cl.**
CPC *H04L 43/0876* (2013.01); *G06Q 30/0206* (2013.01); *H04L 12/1489* (2013.01); *H04L 41/0896* (2013.01); *H04L 41/0897* (2022.05); *H04L 47/12* (2013.01)

(72) Inventors: **Michael Love**, Marble Falls, TX (US);
Blake Love, Austin, TX (US); **Tiago Soromenho**, Austin, TX (US)

(21) Appl. No.: **19/211,683**

(22) Filed: **May 19, 2025**

Related U.S. Application Data

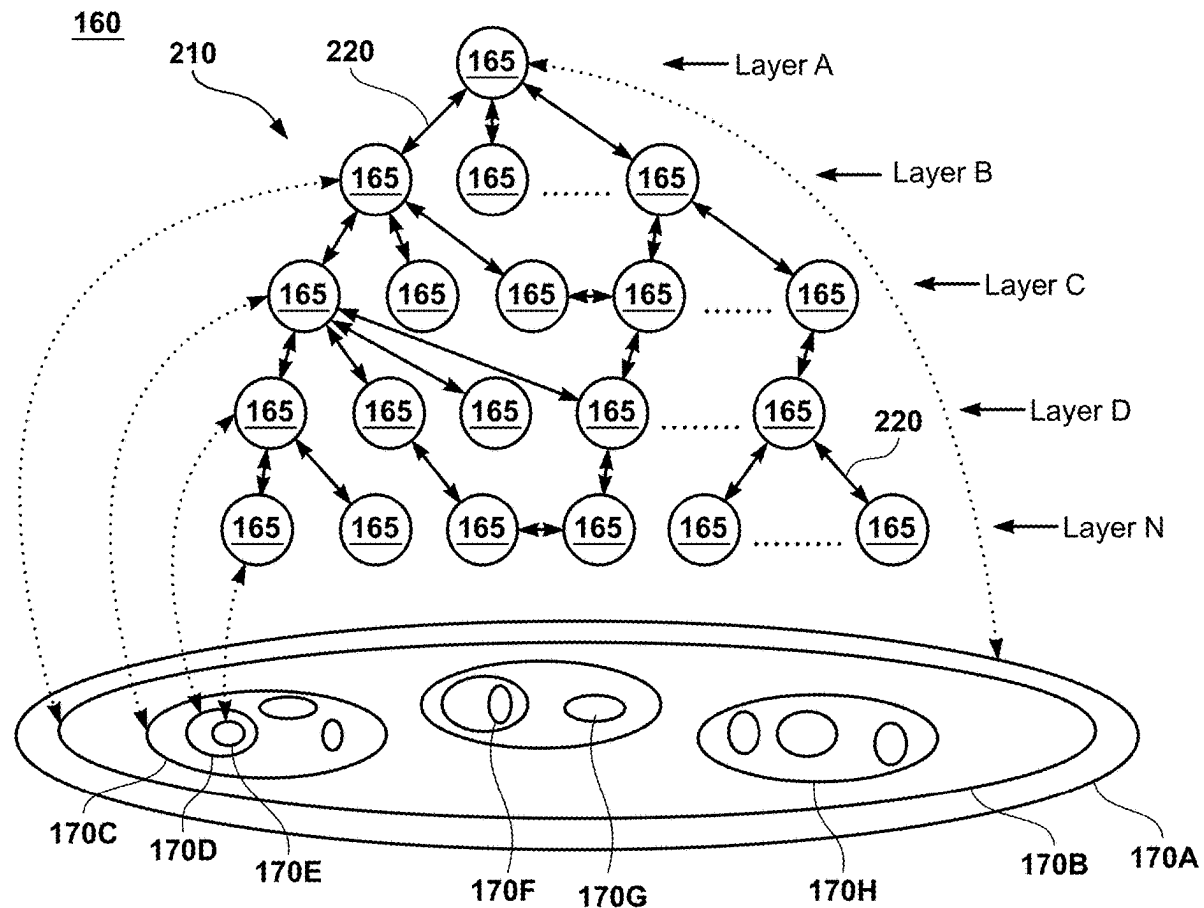
(63) Continuation of application No. 18/136,043, filed on Apr. 18, 2023, now Pat. No. 12,335,122, which is a continuation of application No. 18/074,418, filed on Dec. 2, 2022, now Pat. No. 11,665,076.

Publication Classification

(51) **Int. Cl.**
H04L 43/0876 (2022.01)
G06Q 30/0201 (2023.01)

(57) **ABSTRACT**

This disclosure relates to decentralized computing networks, architectures and techniques for collecting, analyzing, and processing data over multiple channels. A decentralized computing network comprises a plurality of computing nodes, each of which is dedicated to analyzing and processing events for a particular channel corresponding to a geographic region. Each node of the decentralized computing network can operate independently to process channel analysis data for a corresponding channel. The decentralized configuration of the nodes enables efficient processing of data collected over large geographic areas, increases the reliability of the system, and facilitates easy scaling of the system. Other embodiments are disclosed herein as well.



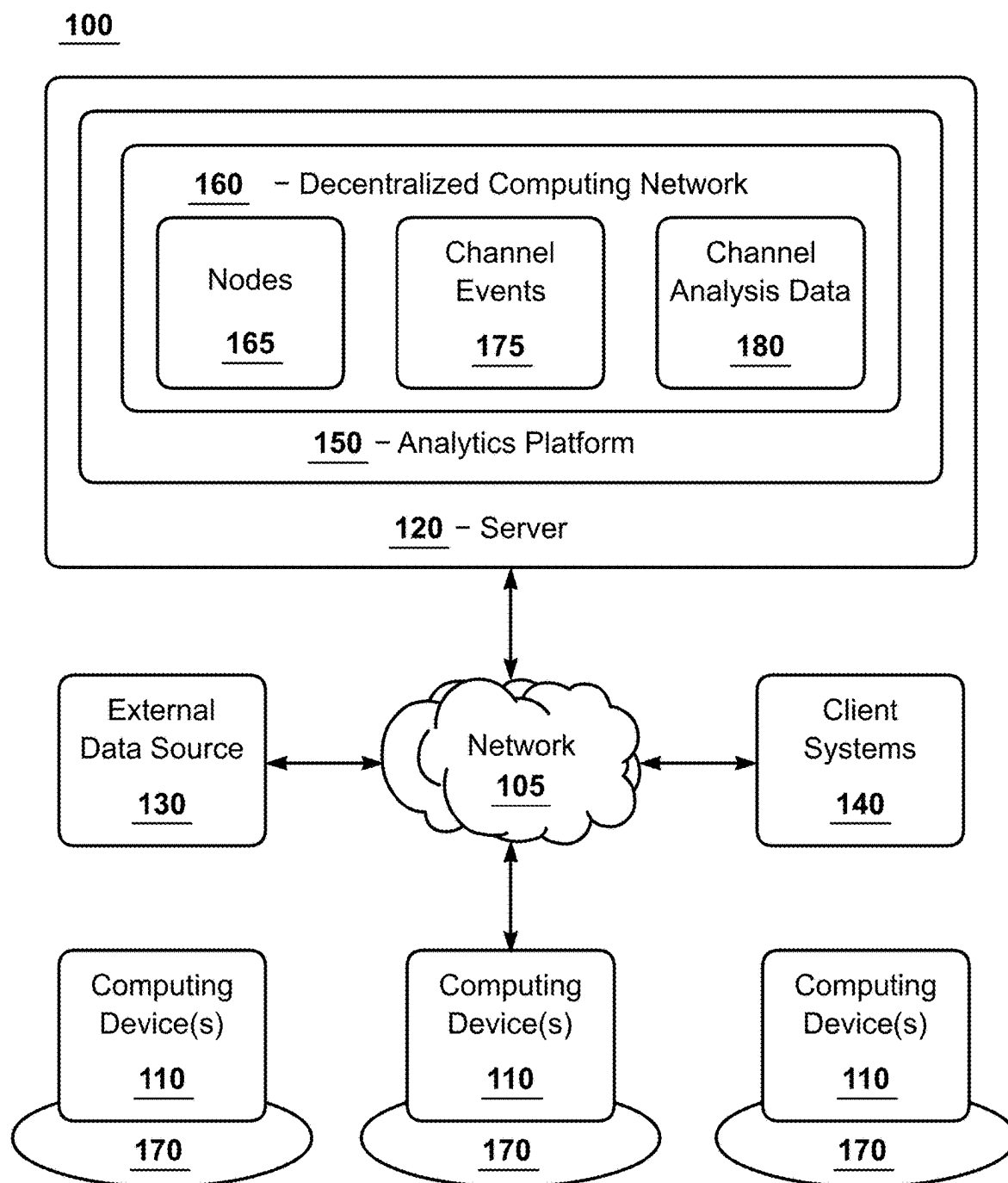


FIG. 1A

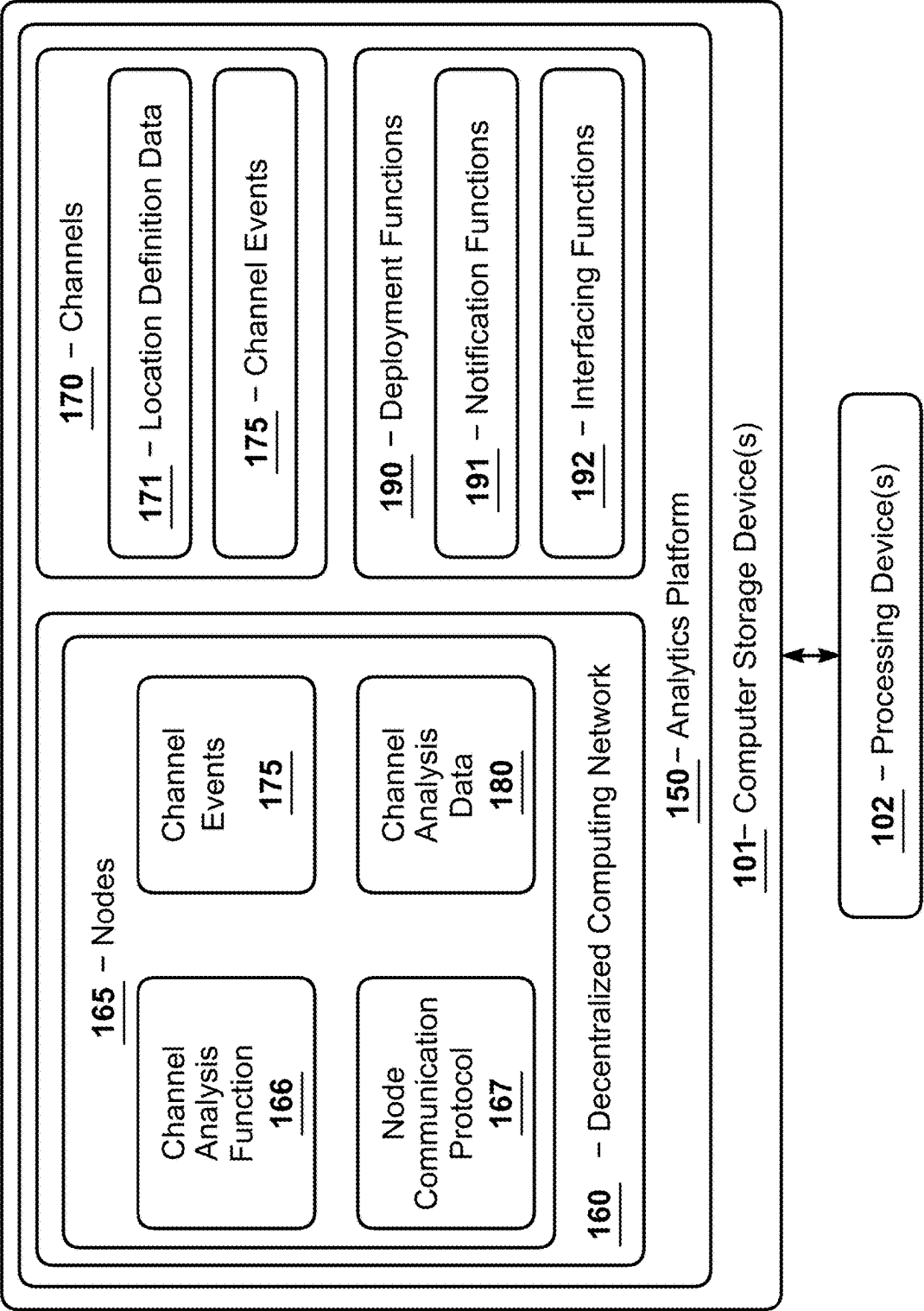


FIG. 1B

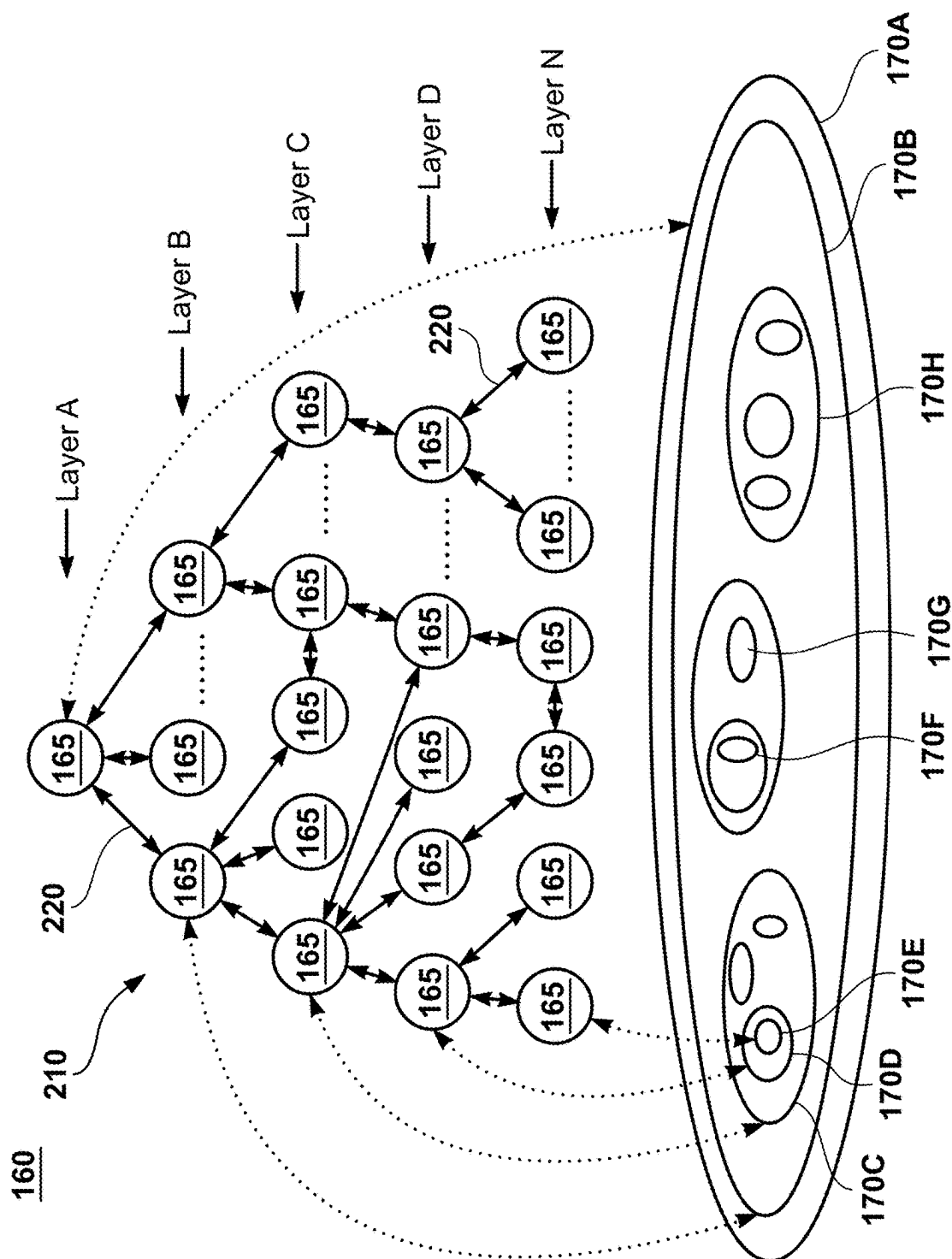


FIG. 2A

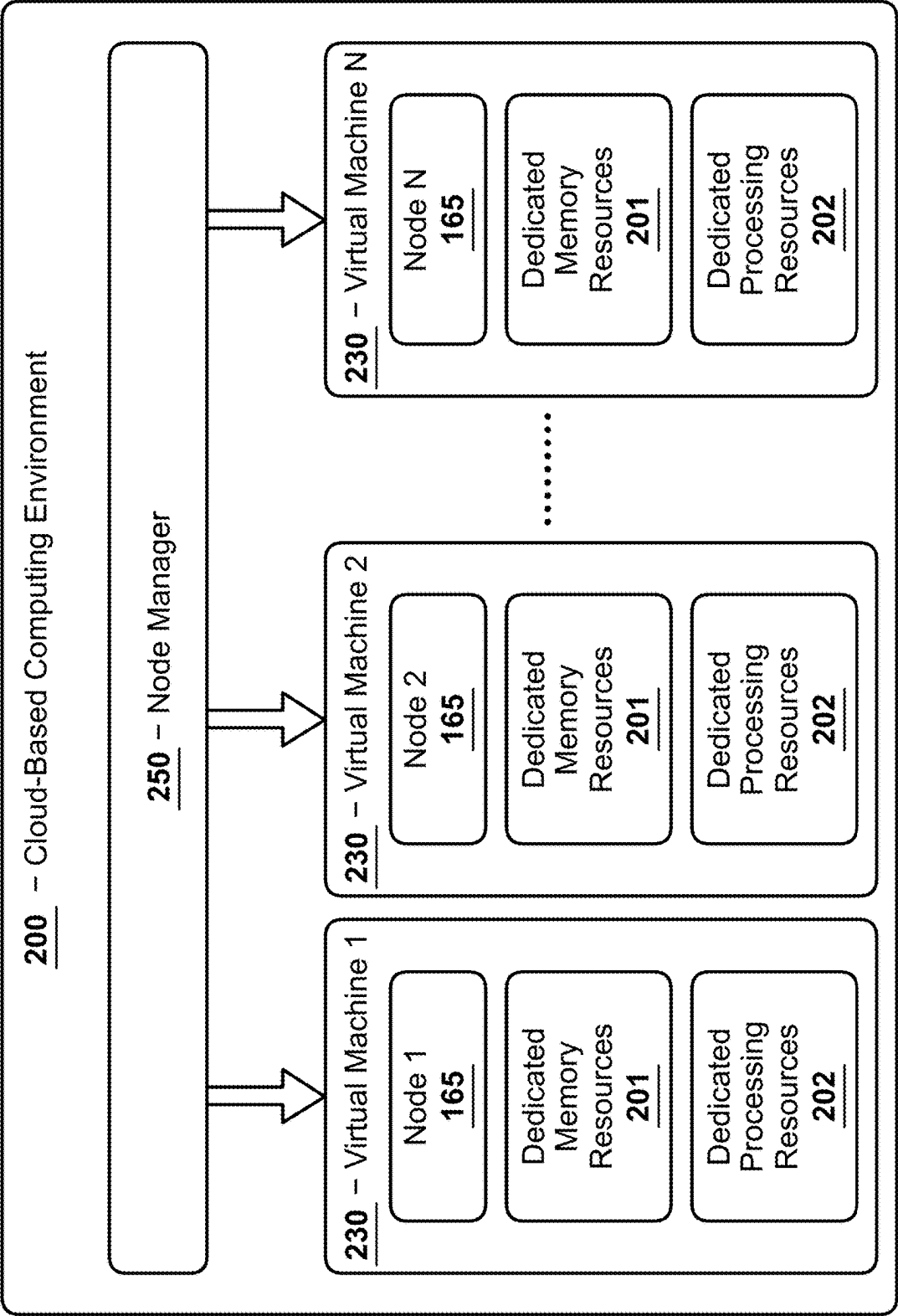


FIG. 2B

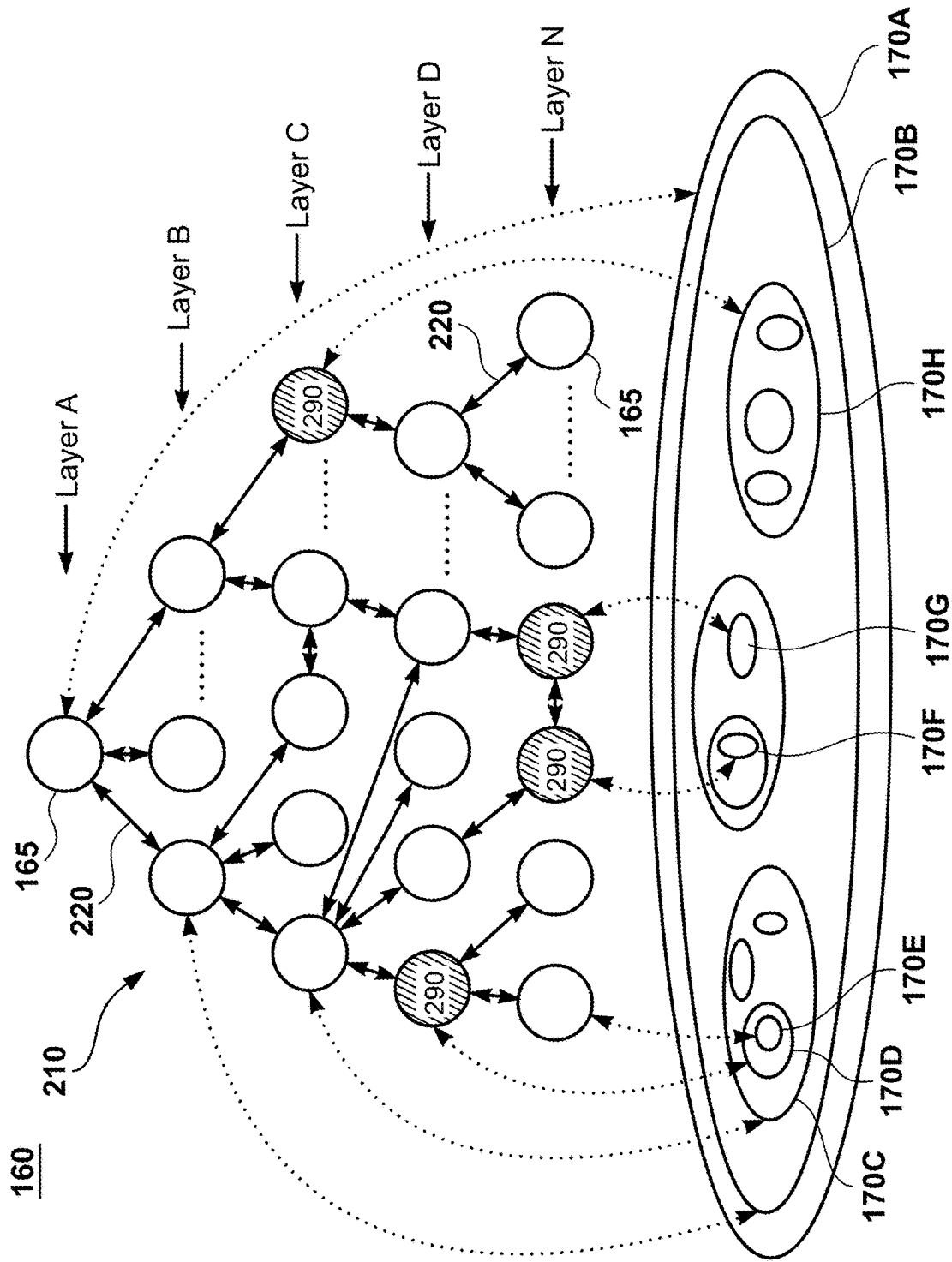


FIG. 2C

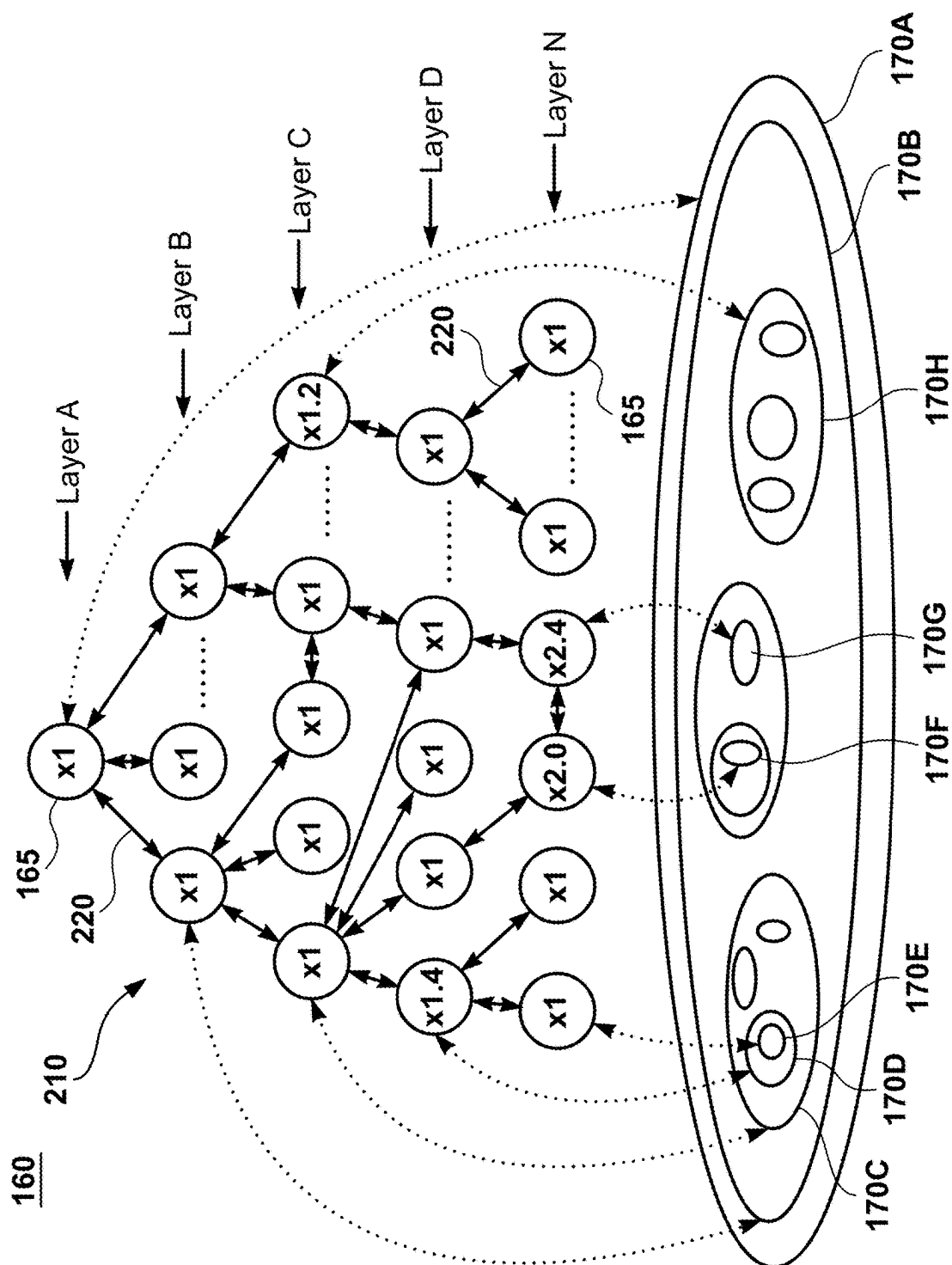


FIG. 2D

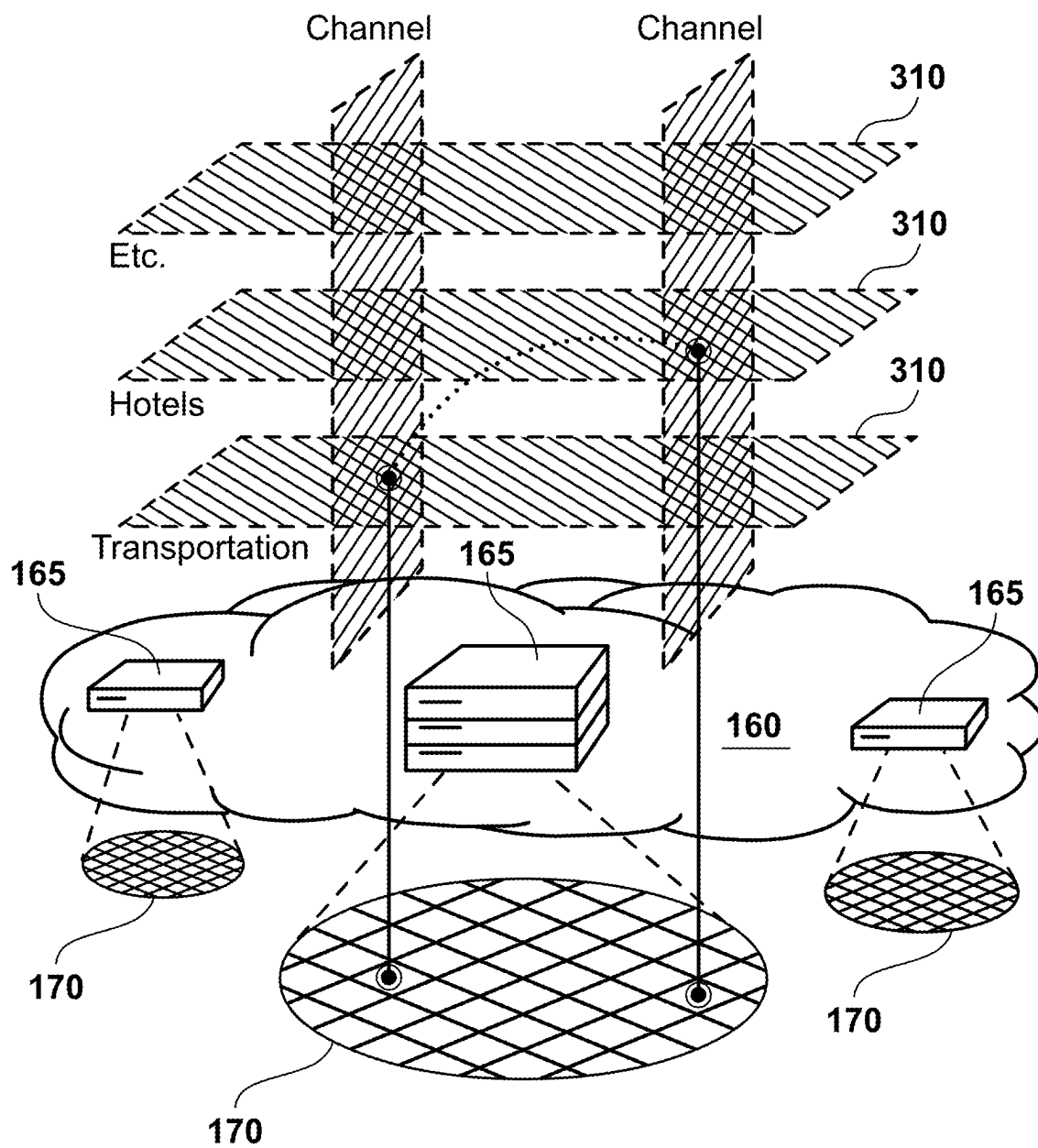


FIG. 3

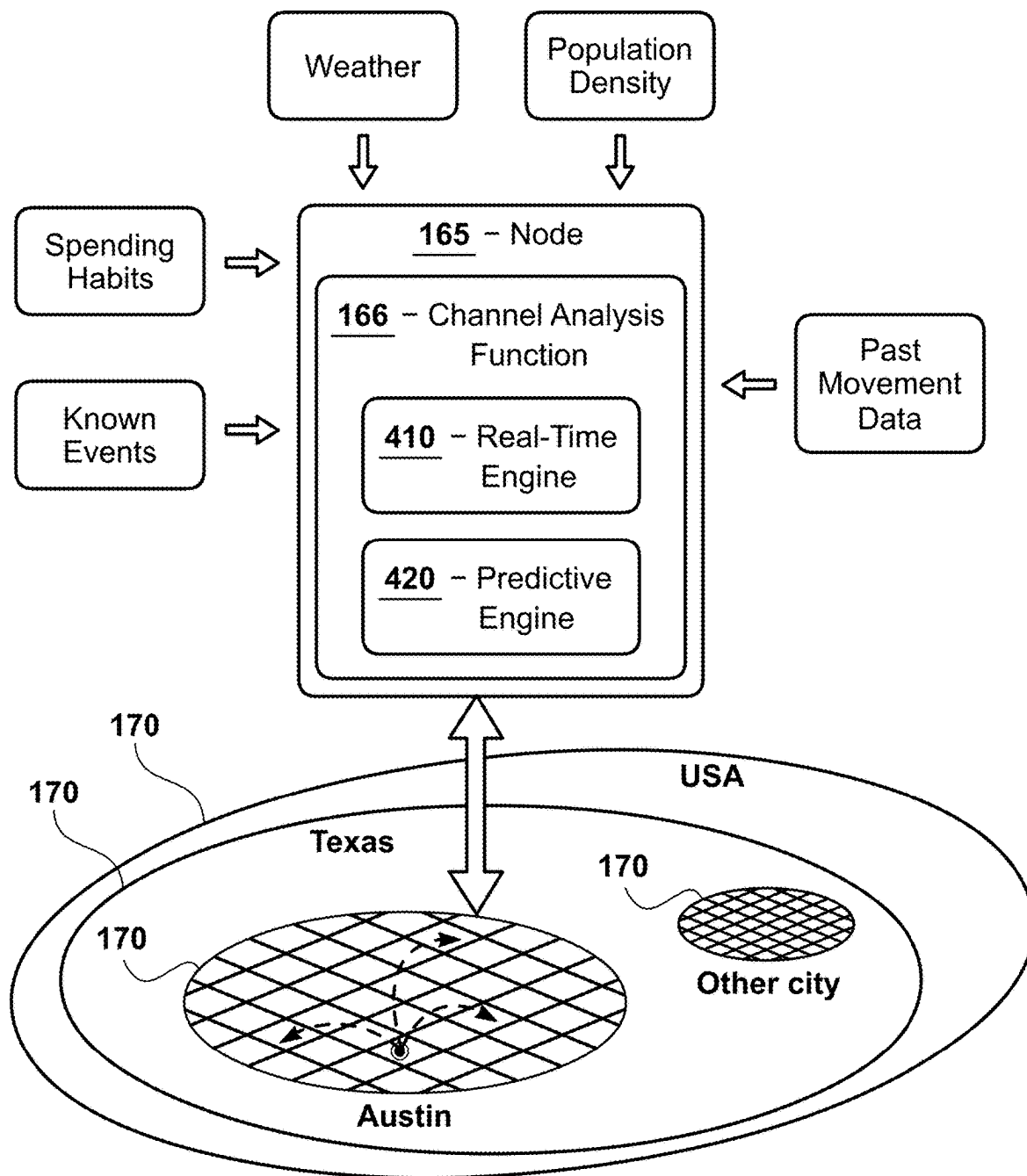


FIG. 4

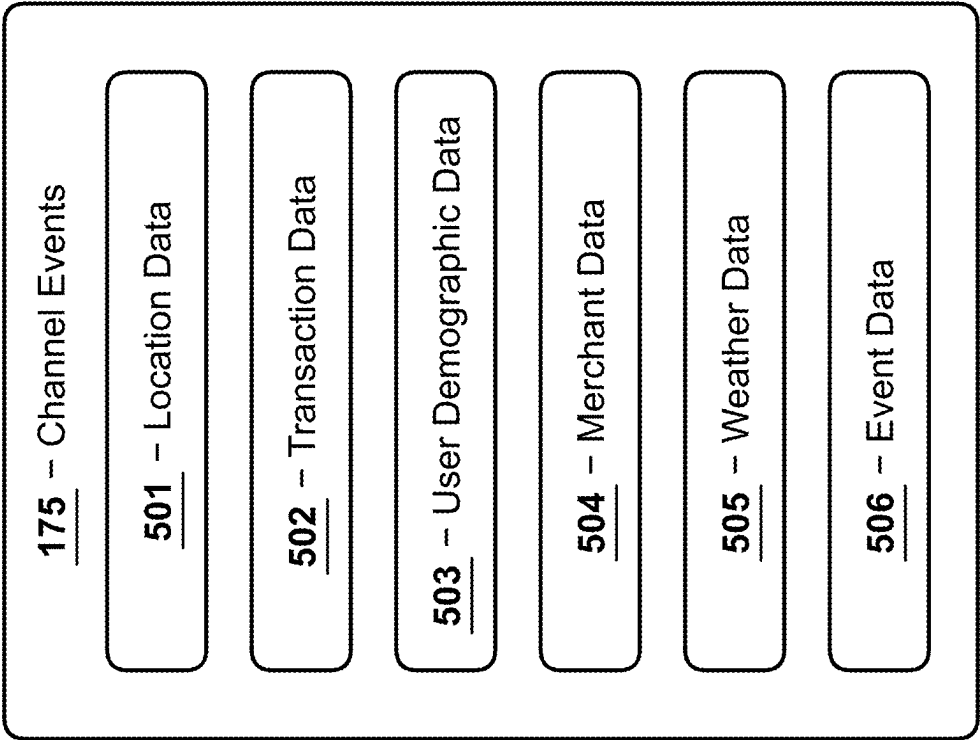


FIG. 5A

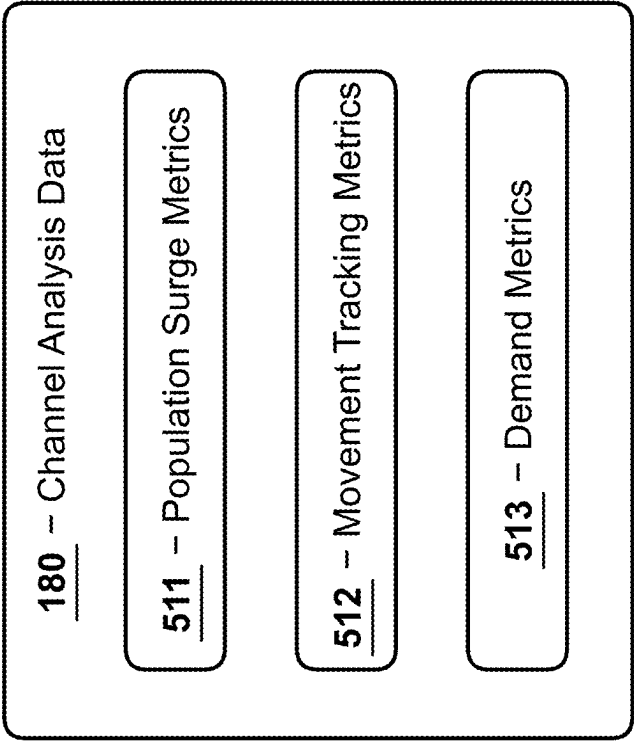


FIG. 5B

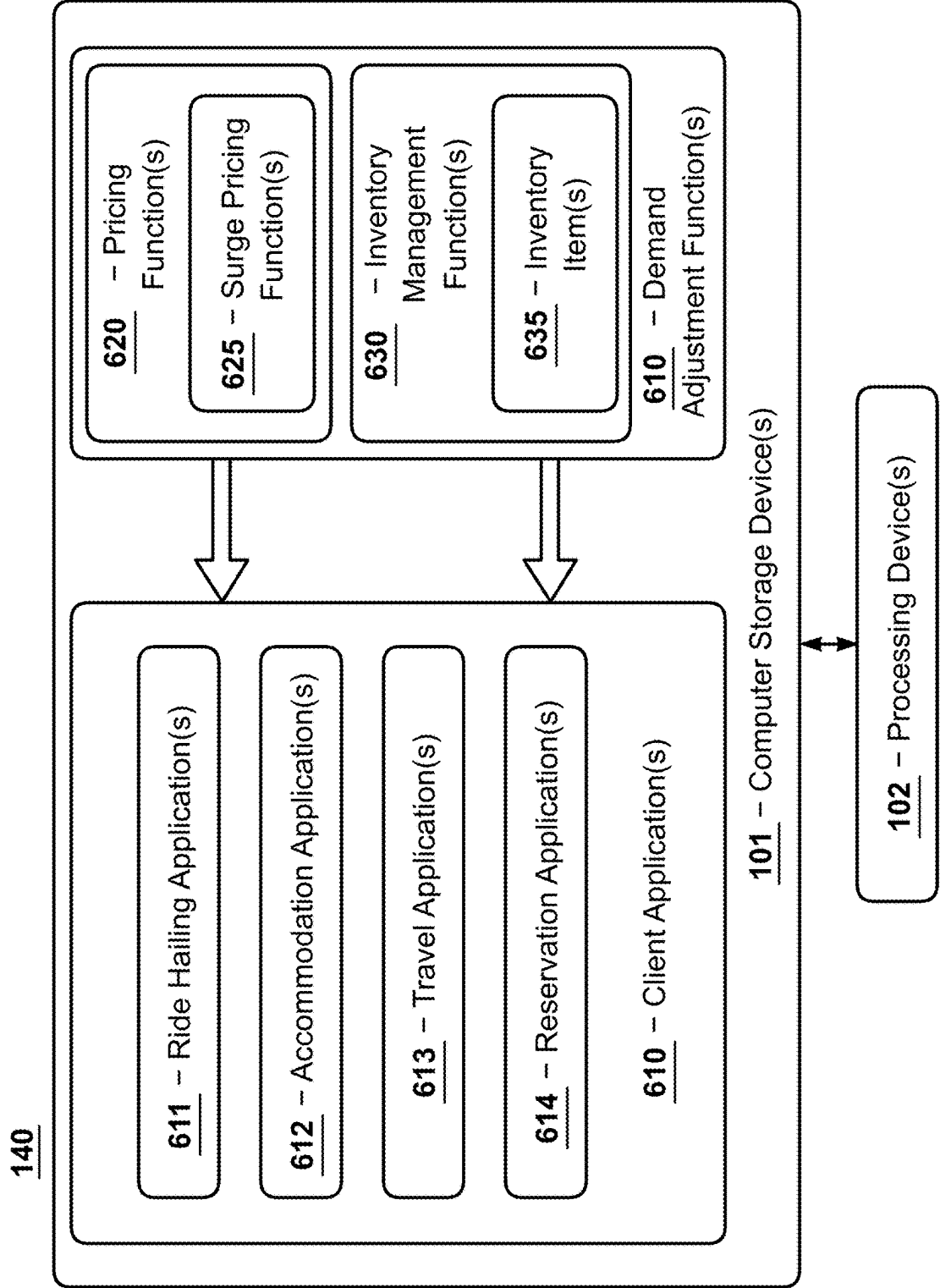


FIG. 6

700

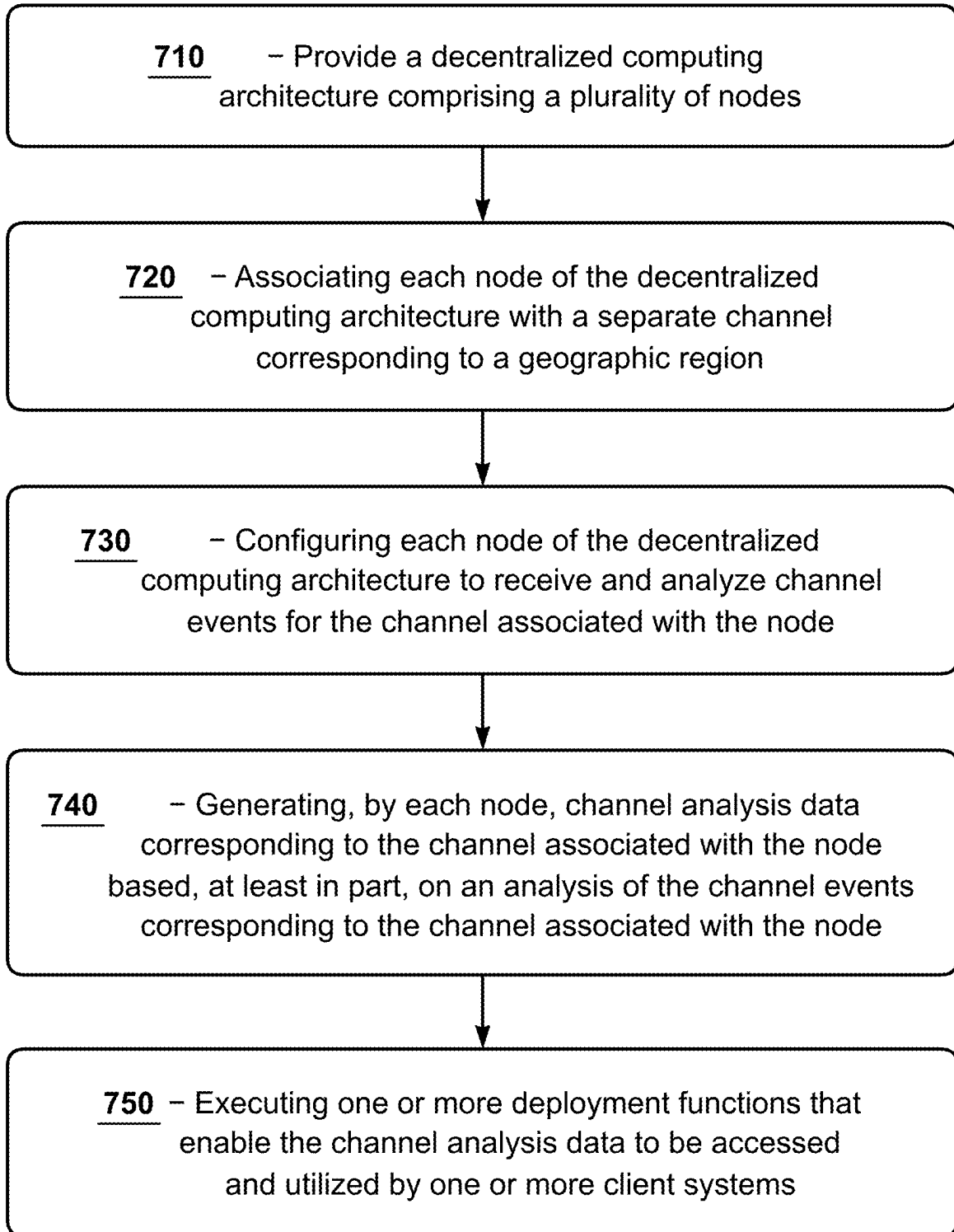


FIG. 7

DECENTRALIZED COMPUTING NETWORKS, ARCHITECTURES AND TECHNIQUES FOR PROCESSING EVENTS ACROSS MULTIPLE CHANNELS

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The present application is a continuation of U.S. patent application Ser. No. 18/136,043 filed on Apr. 18, 2023, which is a continuation of U.S. patent application Ser. No. 18/074,418 filed on Dec. 2, 2022. The contents of the aforementioned applications are herein incorporated by reference in their entirety.

TECHNICAL FIELD

[0002] This disclosure is related to decentralized computing networks, architectures, and techniques for processing events across multiple channels corresponding to distinct geographic regions, as well as systems, methods, and techniques for the same.

BACKGROUND

[0003] Traditional systems for monitoring or analyzing data across large geographic areas typically utilize a centralized network configuration. With a centralized network configuration, all data is collected for analysis and processing by a single, central authority (e.g., a central server) that manages data storage, communications, and data processing operations across the network. For example, in a typical scenario, a central authority may aggregate data across large geographic areas, separately analyze the data for each of the geographic areas, and communicate updates or instructions to devices across the network. Hence, all analysis and decision making across the geographic areas is performed by the central authority.

[0004] Utilizing a centralized network configuration to collect, analyze, and process data across large geographic areas can be disadvantageous for a variety of reasons. One disadvantage relates to that fact that processing aggregated data across large geographic areas with a single central authority can be computationally expensive, and can require large allocations of network storage and processing resources. Moreover, processing large collections of data in this manner can result in network latency, which can be attributed, at least in part, to the time required to process the data and communicate with devices over the network. In scenarios in which systems are intended to provide real-time data or functionality, these latency issues can significantly affect performance of the system and can prevent updates or other data from being disseminated across the network in a timely fashion.

[0005] Another disadvantage of traditional, centralized systems relates to their lack of scalability and flexibility. Centralized systems can be difficult to scale past a certain point because doing so typically requires additional storage, bandwidth, and/or processing power to be provided to the central authority. This can be particularly problematic in scenarios in which a system is rapidly expanding to cover new geographic areas, or in scenarios in which data-intensive upgrades are desired to improve the functionality of the system in existing geographic areas.

[0006] A further disadvantage of centralized systems can be attributed to the fact that the central authority serves as a

single point of failure. If the central authority fails (either for technical reasons or due malicious cyberattacks), the entire system or network also will fail and become unavailable.

BRIEF DESCRIPTION OF DRAWINGS

[0007] To facilitate further description of the embodiments, the following drawings are provided, in which like references are intended to refer to like or corresponding parts, and in which:

[0008] FIG. 1A is a diagram of an exemplary system according to certain embodiments;

[0009] FIG. 1B is a block diagram illustrating exemplary features of surge analytics platform according to certain embodiments;

[0010] FIG. 2A is a diagram illustrating an exemplary geographical hierarchy of nodes for a decentralized computing network according to certain embodiments;

[0011] FIG. 2B is a block diagram illustrating a cloud-based computing environment for a decentralized computing network according to certain embodiments;

[0012] FIG. 2C is a diagram illustrating a decentralized computing network that is configured to execute a demand adjustment function according to certain embodiments;

[0013] FIG. 2D is a diagram illustrating a decentralized computing network that is configured to execute a demand adjustment function according to certain embodiments;

[0014] FIG. 3 is an illustration demonstrating how a decentralized computing network can generate channel analysis data across various verticals according to certain embodiments;

[0015] FIG. 4 is an illustration demonstrating aspects of a predictive engine according to certain embodiments;

[0016] FIG. 5A is a block diagram illustrating exemplary channel events according to certain embodiments;

[0017] FIG. 5B is a block diagram illustrating exemplary channel analysis data according to certain embodiments;

[0018] FIG. 6 is a block diagram of an exemplary client system according to certain embodiments; and

[0019] FIG. 7 is a flowchart illustrating an exemplary method according to certain embodiments.

[0020] The terms “first,” “second,” “third,” “fourth,” and the like in the description and in the claims, if any, are used for distinguishing between similar elements and not necessarily for describing a particular sequential or chronological order. It is to be understood that the terms so used are interchangeable under appropriate circumstances such that the embodiments described herein are, for example, capable of operation in sequences other than those illustrated or otherwise described herein.

[0021] The terms “upper,” “lower,” “left,” “right,” “front,” “rear,” “back,” “top,” “bottom,” “over,” “under,” and the like in the description and in the claims, if any, are used for descriptive purposes and not necessarily for describing permanent relative positions. It is to be understood that the terms so used are interchangeable under appropriate circumstances such that the embodiments of the systems, methods, and/or articles of manufacture described herein are, for example, capable of operation in other orientations than those illustrated or otherwise described herein.

DETAILED DESCRIPTION OF EXEMPLARY EMBODIMENTS

[0022] The present disclosure relates to systems, methods, apparatuses, and computer program products that utilize decentralized computing networks, architectures and techniques to collect, analyze, and process data over multiple geographic areas. In certain embodiments, a decentralized computing network comprises a plurality of computing nodes, each of which is configured to collect, analyze, and process channel events for a channel that corresponds to a particular geographic region. Each node of the decentralized computing network can operate independently to process data originating from, or associated with, a given channel. For example, each node can receive channel events originating from, or associated with, a given channel, and can execute a channel analysis function to derive channel analysis data pertaining to the channel or its corresponding geographic region. As described in further detail below, the channel analysis data generated by each node can be used for various purposes. The decentralized configuration of the nodes enables efficient processing of data collected over large geographic areas, increases the reliability of the system, and facilitates easy scaling of the system.

[0023] In certain embodiments, the nodes included in the decentralized computing network can be arranged in a geographical hierarchy that is based on the channels or geographic areas associated with the nodes. The geographical hierarchy can establish parent-child relationships among the nodes, such that each child node corresponds to a geographic region that is a subset of a geographic region associated with a parent node. While each node can operate independently to process data for a corresponding channel or geographic region, a node communication protocol can enable information to be exchanged among the nodes based on relationships of the nodes to each other. For example, in some cases, a given node can exchange information with one or more parent nodes and/or one or more child nodes based on geographical associations or relationships. The exchange of information among the nodes avoids redundancies, reduces storage requirements, and increases efficiency with respect to generating the channel analysis data.

[0024] The types of channel events received and processed by each node can vary. The channel events can generally include any data relating to individuals located within a channel, activities occurring within the channel, and/or other conditions associated with the channel. For example, in some embodiments, the channel events received by a node for a given channel can include data indicating locations or movements of individuals (or devices operated by individuals) within the channel. The channel events also can include data relating to transactions conducted within the channel, weather conditions within the channel, merchants located within the channel, and events (e.g., concerts, conventions, sporting games, etc.) occurring with the channel. The channel events can include many additional attributes related to the channel itself and/or users within the channel. Further examples of channel events are described throughout this disclosure.

[0025] Each node of the decentralized computing network can execute a channel analysis function that is configured to analyze or process the channel events, and generate channel analysis data for a corresponding channel associated with the node. The channel analysis data generated by each node can vary. The channel analysis data for a given channel can

include various metrics relating to the channel itself, individuals within the channel, and/or activities occurring within the channel. For example, the channel analysis data can indicate or predict the supply and/or demand within the channel for one or more products or services (e.g., riding sharing, hotel accommodations, parking garages, dining availability, sporting or event tickets, etc.). The channel analysis data also can identify or predict increases and/or decreases in the number of individuals within the channel, as well as intra-channel and inter-channel movements of those individuals. The channel analysis data can include many other metrics related to the channel, individuals in the channel, and/or activities occurring within the channel. Further examples of channel analysis data are described throughout this disclosure.

[0026] The channel analysis data generated by each node can include real-time analysis information that provides current or up-to-date data relating to the aforementioned metrics and/or other metrics. The channel analysis data also can include predictive analysis information that predicts a future status of the aforementioned metrics and/or other metrics. Examples of these real-time and predictive analytics are provided throughout this disclosure.

[0027] In certain embodiments, each of the nodes can utilize the channel analysis data to execute one or more demand adjustment functions. Additionally, or alternatively, the channel analysis data can be provided to one or more client systems that utilize the channel analysis data to execute the one or more demand adjustment functions. One exemplary demand adjustment function can include a pricing function that determines prices for one or more inventory items. For example, in some cases, the channel analysis data can be utilized by a surge pricing function to dynamically adjust prices for one or more inventory items in a manner that accounts for the supply and demand for the one or more inventory items. Another exemplary demand adjustment function includes an inventory management function that utilizes the channel analysis data to manage or adjust inventories (e.g., such as to dynamically reallocate inventory items among channels and/or initiate ordering of additional inventory items).

[0028] The channel analysis data generated by a given node can be utilized to execute one or more deployment functions that enable the channel analysis data to be utilized for a variety of purposes. In some scenarios, a deployment function can be executed to automate pricing systems (e.g., surge pricing systems) and/or inventory systems based on the channel analysis data. For example, the channel analysis data can be interfaced with pricing and inventory applications operated by one or more client systems (e.g., third-party systems) to automatically adjust the pricing of inventory (e.g., products or services) and/or automatically reallocate or reorder inventory based on the supply and demand metrics associated with the channel and/or based on the increases or decreases of individuals within the channel.

[0029] In another example, a deployment function can be executed to transmit notifications that include some or all of the channel analysis data. For example, notifications can be transmitted to client systems affiliated with merchants within the channel to disseminate real-time data and/or predictive data relating to the supply and demand of particular products or services. The merchant notifications also can include other information included in the channel analysis data (e.g., population surge information, event information, transaction

patterns within the channel, movement patterns of individuals with the channel, etc.). Notifications also can be transmitted to individuals within the channel (e.g., to present offers or discounts to the individuals and/or recommend merchants located in the channel). Additional types of deployment functions are described throughout this disclosure.

[0030] The technologies described herein provide a variety of benefits and advantages. Amongst other things, the decentralized computing techniques enable rapid and efficient processing of data collected over multiple geographic areas. This can be attributed, at least in part, to the decentralization of nodes that act independently to collect and process information for specific channels, rather than relying on a central authority to aggregate and analyze the data across multiple geographic areas. Another benefit of the centralized computing techniques relates to increases the reliability of the system. In the event that one of the nodes fail, the other nodes can continue to function independently. Additionally, the failed node easily can be replicated and reinitialized to minimize downtime. Another benefit relates to the ease of scaling of the system, such as to accommodate new geographic areas and/or enhanced functionality. When geographic scaling is desired, additional nodes can easily be incorporated into the geographical hierarchy without having to modify the functionality of a central authority. Likewise, when enhanced functionality is desired, additional computing and storage resources can easily be allocated to existing nodes.

[0031] Further benefits can be attributed to the usefulness and versatility of the channel analysis data, which can be leveraged for many different purposes. This data can provide merchant entities with real-time and/or predictive metrics that can assist merchant entities with accommodating surges in supply and demand (and/or population surges within a channel). In some cases, the channel analysis data also can be integrated with pricing and/or inventory systems to seamlessly adjust pricing and/or inventory allocations in real-time or near real-time.

[0032] Additional benefits can be attributed to embodiments that utilize the channel analysis data to automated surge pricing functions. Client applications that employ surge pricing functionalities can better mitigate imbalances between an available supply of inventory items and a demand for those inventory items. The channel analysis can be leveraged to dynamically adjust prices for the inventory items, thereby enabling providers of the client applications to reduce high-demand peaks.

[0033] The technologies discussed herein can be used in a variety of different contexts and environments. Some useful applications of these technologies are in the context of adjusting pricing and/or inventory allocations for merchant entities, such as those that provide ride hailing services, hotel accommodations, parking garages, bar and restaurant establishments, etc. For example, the technologies disclosed herein can provide merchant-specific insights and metrics regarding the current and future demand for products and services offered by these entities. These metrics can be used by those entities to automatically or manually adjust settings for pricing systems (e.g., surge pricing systems), inventory systems, and/or other operations.

[0034] The embodiments described in this disclosure can be combined in various ways. Any aspect or feature that is described for one embodiment can be incorporated to any

other embodiment mentioned in this disclosure. Moreover, any of the embodiments described herein may be hardware-based, may be software-based, or, preferably, may comprise a mixture of both hardware and software elements. Thus, while the description herein may describe certain embodiments, features, or components as being implemented in software or hardware, it should be recognized that any embodiment, feature and/or component referenced in this disclosure can be implemented in hardware and/or software.

[0035] FIG. 1A is a diagram of an exemplary system **100** in accordance with certain embodiments. The system **100** includes, inter alia, an analytics platform **150** that utilizes a decentralized computing network **160** to generate or derive channel analysis data **180** for a plurality of channels **170**. FIG. 1B is a diagram illustrating additional features, components, and/or functions associated with the analytics platform **150** and decentralized computing network **160**. FIGS. 1A and 1B are jointly discussed below.

[0036] The system **100** comprises one or more computing devices **110**, one or more servers **120**, one or more external data sources **130**, and one or more client systems **140** that are in communication over a network **105**. An analytics platform **150** comprising a decentralized computing network **160** is stored on, and executed by, the one or more servers **120**. The network **105** may represent any type of communication network, e.g., such as one that comprises a local area network (e.g., a Wi-Fi network), a personal area network (e.g., a Bluetooth network), a wide area network, an intranet, the Internet, a cellular network, a television network, and/or other types of networks.

[0037] All the components illustrated in FIG. 1A, including the one or more computing devices **110**, one or more servers **120**, one or more external data sources **130**, and one or more client systems **140**, and analytics platform **150** can be configured to communicate directly with each other and/or over the network **105** via wired or wireless communication links, or a combination of the two. Each of the computing devices **110**, servers **120**, external data sources **130**, client systems **140**, and analytics platform **150** can include one or more communication devices, one or more computer storage devices **101**, and one or more processing devices **102** (FIG. 1B) that are capable of executing computer program instructions.

[0038] The one or more computer storage devices **101** may include (i) non-volatile memory, such as, for example, read only memory (ROM) and/or (ii) volatile memory, such as, for example, random access memory (RAM). The non-volatile memory may be removable and/or non-removable non-volatile memory. Meanwhile, RAM may include dynamic RAM (DRAM), static RAM (SRAM), etc. Further, ROM may include mask-programmed ROM, programmable ROM (PROM), one-time programmable ROM (OTP), erasable programmable read-only memory (EPROM), electrically erasable programmable ROM (EEPROM) (e.g., electrically alterable ROM (EAROM) and/or flash memory), etc. In certain embodiments, the one or more computing storage devices **101** may be physical, non-transitory mediums. The one or more computer storage devices **101** can store, inter alia, instructions associated the implementing the functionalities of the analytics platform **150** and decentralized computing network **160** described herein.

[0039] The one or more processing devices **102** may include one or more central processing units (CPUs), one or more microprocessors, one or more microcontrollers, one or

more controllers, one or more complex instruction set computing (CISC) microprocessors, one or more reduced instruction set computing (RISC) microprocessors, one or more very long instruction word (VLIW) microprocessors, one or more graphics processor units (GPU), one or more digital signal processors, one or more application specific integrated circuits (ASICs), and/or any other type of processor or processing circuit capable of performing desired functions. The one or more processing devices **102** can be configured to execute any computer program instructions that are stored or included on the one or more computer storage devices **101** including, but not limited to, instructions associated with the implementing the functionalities of the analytics platform **150** and decentralized computing network **160** described throughout this disclosure.

[0040] Each of the one or more communication devices can include wired and wireless communication devices and/or interfaces that enable communications using wired and/or wireless communication techniques. Wired and/or wireless communication can be implemented using any one or combination of wired and/or wireless communication network topologies (e.g., ring, line, tree, bus, mesh, star, daisy chain, hybrid, etc.) and/or protocols (e.g., personal area network (PAN) protocol(s), local area network (LAN) protocol(s), wide area network (WAN) protocol(s), cellular network protocol(s), powerline network protocol(s), etc.). Exemplary PAN protocol(s) can comprise Bluetooth, Zigbee, Wireless Universal Serial Bus (USB), Z-Wave, etc. Exemplary LAN and/or WAN protocol(s) can comprise Institute of Electrical and Electronic Engineers (IEEE) 802.3 (also known as Ethernet), IEEE 802.11 (also known as WiFi), etc. Exemplary wireless cellular network protocol(s) can comprise Global System for Mobile Communications (GSM), General Packet Radio Service (GPRS), Code Division Multiple Access (CDMA), Evolution-Data Optimized (EV-DO), Enhanced Data Rates for GSM Evolution (EDGE), Universal Mobile Telecommunications System (UMTS), Digital Enhanced Cordless Telecommunications (DECT), Digital AMPS (IS-136/Time Division Multiple Access (TDMA)), Integrated Digital Enhanced Network (iDEN), Evolved High-Speed Packet Access (HSPA+), Long-Term Evolution (LTE), WiMAX, etc. The specific communication software and/or hardware can depend on the network topologies and/or protocols implemented. In certain embodiments, exemplary communication hardware can comprise wired communication hardware including, but not limited to, one or more data buses, one or more universal serial buses (USBs), one or more networking cables (e.g., one or more coaxial cables, optical fiber cables, twisted pair cables, and/or other cables). Further exemplary communication hardware can comprise wireless communication hardware including, for example, one or more radio transceivers, one or more infrared transceivers, etc. Additional exemplary communication hardware can comprise one or more networking components (e.g., modulator-demodulator components, gateway components, etc.). In certain embodiments, the one or more communication devices can include one or more transceiver devices, each of which includes a transmitter and a receiver for communicating wirelessly. The one or more communication devices also can include one or more wired ports (e.g., Ethernet ports, USB ports, auxiliary ports, etc.) and related cables and wires (e.g., Ethernet cables, USB cables, auxiliary wires, etc.).

[0041] In certain embodiments, the one or more communication devices additionally, or alternatively, can include one or more modem devices, one or more router devices, one or more access points, and/or one or more mobile hot spots. For example, modem devices may enable some or all of the computing devices **110**, servers **120**, external data sources **130**, client systems **140**, and/or analytics platform **150** to be connected to the Internet and/or other network. The modem devices can permit bi-directional communication between the Internet (and/or other network) and the computing devices **110**, servers **120**, external data sources **130**, client systems **140**, and/or analytics platform **150**. In certain embodiments, one or more router devices and/or access points may enable the computing devices **110**, servers **120**, external data sources **130**, client systems **140**, and/or analytics platform **150** to be connected to a LAN and/or other more other networks. In certain embodiments, one or more mobile hot spots may be configured to establish a LAN (e.g., a Wi-Fi network) that is linked to another network (e.g., a cellular network). The mobile hot spot may enable the computing devices **110**, servers **120**, external data sources **130**, client systems **140**, and/or analytics platform **150** to access the Internet and/or other networks.

[0042] In certain embodiments, the computing devices **110** may represent desktop computers, laptop computers, mobile devices (e.g., smart phones, personal digital assistants, tablet devices, vehicular computing devices, wearable devices, or any other device that is mobile in nature), and/or other types of devices. The one or more servers **120** may generally represent any type of computing device, including any of the aforementioned computing devices **110**. The one or more servers **120** also can comprise one or more mainframe computing devices and/or one or more virtual servers that are executed in a cloud-computing environment. In some embodiments, the one or more servers **120** can be configured to execute web servers and can communicate with the computing devices **110**, external data sources **130**, client systems **140**, and/or other devices over the network **105** (e.g., over the Internet).

[0043] In certain embodiments, the analytics platform **150** can be stored on, and executed by, the one or more servers **120**. Additionally, or alternatively, the analytics platform **150** can be stored on, and executed by, the one or more computing devices **110** and/or one or more client systems **140**. The analytics platform **150** can be executed be stored on, and executed, by other devices as well.

[0044] In some embodiments, the analytics platform **150** also can be stored as a local application on a computing device **110**, or interfaced with a local application stored on a computing device **110**, to implement the techniques and functions described herein. The computing device **110** may be part of client system **140** in some scenarios.

[0045] The client systems **140** can generally correspond to third-party systems, networks, and/or devices that access the analytics platform **150** and/or utilize the data (including the channel analysis data **180**) generated by the analytics platform **150**. For example, the client systems **140** can be operated and managed by individuals, businesses, and/or other entities that utilize the analytics platform **150** (including the channel analysis data **180** generated by the analytics platform **150**) to improve the functionalities of one or more systems and/or one or more applications.

[0046] In certain embodiments, each of the client systems **150** can register and/or create a user account with the

analytics platform **150** to obtain access to the data and services provided by the analytics platform **150**. The client systems **180** can be operated by, or associated with, individuals or businesses from any industry or vertical including, but not limited, to those that offer ride hailing services, hotel or lodging accommodations, parking space availability, tavern services, and restaurant services. As explained in further detail below, the client systems **180** can utilize the channel analysis data **180** (and other data provided by the analytics platform **150**) to enhance and improve business operations in various ways.

[0047] Each of the client systems **140** may include one or more computing devices **110** that enable the client systems **140** to access the analytics platform **150** over the network **105**. In some cases, one or more of the client systems **140** may include sophisticated technological infrastructures, such those that include enterprise systems, servers **120**, virtual private networks (VPNs), intranets, etc. The computing devices **110**, servers **120**, and/or other devices associated with each client system **140** can store and execute various applications (e.g., such as ride hailing applications, lodging booking applications, dining reservation applications, pricing applications, inventory management applications, etc.). The client systems **140** and associated applications can leverage the data (e.g., channel analysis data **180**) provided by the analytics platform **150** in various ways.

[0048] In certain embodiments, the analytics platform **150** can be integrated with (or can communicate with) various applications hosted by the client systems **140** including, but not limited to, applications that provide products or services for transportation services (e.g., ride hailing services, ride sharing services, vehicle rental services, and/or ticket scheduling services for buses, trains, planes, boats, and/or other modes of transportation), lodging accommodations (e.g., booking services for hotels, motels, short-term home stays, rental services, property purchases, etc.), parking space services (e.g., booking services for parking garages, parking lots, etc.), and scheduling services (e.g., reservation services for restaurants, bars, sporting events, concerts, ticketed events, etc.). In certain embodiments, the analytics platform **150** additionally, or alternatively, can be integrated with (or can communicate with) e-commerce applications, pricing applications, inventory management applications, and/or other applications.

[0049] The aforementioned applications and/or other applications, each of which may be integrated or interfaced with the analytics platform **150**, can be stored on one or more client systems **140** in some embodiments. For example, the aforementioned applications and/or other applications can be stored on one or more computing device **110** and/or one or more servers **120** associated with one or more client systems **140**.

[0050] As discussed throughout this disclosure, the analytics platform **150** can generally provide functions associated with analyzing conditions associated with activities, individuals, events, and/or like within a plurality of channels **170**. This analysis can be used to generate channel analysis data **180** corresponding to each of the channels **170**. The channel analysis data **180** for a given channel **170** can include various metrics and information useful for understanding and/or predicting the conditions within the channel **170**. For example, the channel analysis data **180** for a channel **170** can include information that indicates and/or predicts population surges within the channel **170**, move-

ments of individuals within the channel, and/or supply and demand for inventory (e.g., products or services) within the channel **170**. Additionally, as described in further detail below, the analytics platform **150** can utilize the channel analysis data **180** to implement a surge pricing function and/or inventory management function on a variety of client applications.

[0051] Each channel **170** can represent, or correspond to, a specific geographic region or area. The scope or region associated with each channel **170** can vary significantly. For example, macro-level channels **170** can correspond to large geographic areas covering entire continents, countries, and/or states. Other more micro-level channels **170** can correspond to counties, cities, and/or towns. Additional channels **170** can correspond to specific regions, neighborhoods, areas, or the like within cities or towns. Regardless of geographic scope, a node **165** included in the decentralized computing network **160** can be associated with each channel **170** to process data associated with the channel **170** and generate channel analysis data **180** for the channel **170**.

[0052] The nodes **165** and/or analytics system **150** can store location definition data **171** that defines the geographic region associated with each channel **170**. For example, in certain embodiments, the location definition data **171** can store global positioning system (GPS) coordinates for each channel **170** that precisely defines the geographic region associated with the channel **170**. Additionally, or alternatively, the location definition data **171** can include geofencing data that defines the geographic region associated with the channel **170**. In some embodiments, the analytics platform **150** can provide access to one or more GUIs that enable users to define the geographic regions associated with the channels **170**.

[0053] The analytics platform **150** can include, or communicate with, a decentralized computing network **160** that is configured to generate some or all of the channel analysis data **180** for the channels **170**. The decentralized computing network **160** can comprise a plurality of nodes **165**, each of which is configured to execute a channel analysis function **166** that independently processes information associated with a dedicated channel **170** and generates corresponding channel analysis data **180** for that channel **170**.

[0054] Each node **175** in the decentralized computing network **160** can be dedicated to a particular channel **170** and can receive channel events **175** associated with the channel **170**. These channel events **175** can be analyzed and/or utilized by the channel analysis function **166** to generate the channel analysis data **180** for the channel **170**.

[0055] Each node **175** of the decentralized computing network **160** can utilize dedicated processing and/or storage resources to process the channel events **175** and generate the channel analysis data **180** for a corresponding channel **170**. In certain embodiments, the decentralized computing network **160** can be provided via a cloud-based computing environment that executes a separate virtual machine for each node **165**. Alternatively, or additionally, one or more hardware servers (or other computing devices) can be dedicated to each node **165**.

[0056] Because the nodes **165** can be configured to separately and independently process data associated each channel **170**, each node **165** only processes the data corresponding to its designated channel **170** and does not need to process the global data across all channels **170**. This decentralized configuration can enable rapid and efficient process-

ing of data collected over multiple geographic areas, and can significantly decrease latencies across the network. The configuration can be particularly beneficial in scenarios in which the analytics platform **150** is collecting and analyzing information over large geographic areas. Additionally, the independent processing of data by the nodes **165** provides more reliability and drastically reduces the chances of a catastrophic system failure in which the entirety of the systems becomes unavailable. Additional details and exemplary configurations of decentralized computing networks **160** are described below with reference to FIGS. 2A and B.

[0057] The types and content of the channel events **175** received and processed by each node **165** can vary. The channel events **175** can generally include any data associated with monitoring locations of individuals within a channel **170**, activities occurring within the channel **1701**, and/or other conditions associated with the channel **170**. For example, the channel events **175** received by a node **165** can include data indicating locations of individuals (or their smart phones or mobile devices) within the channel **170**, transactions conducted within the channel **170**, weather conditions within the channel, and/or events (e.g., concerts, conventions, etc.) occurring within the channel **170**.

[0058] The channel events **175** can be generated by, or received from, various devices, systems, and/or sources. Some of the channel events **175** can be generated by computing devices **110** (e.g., mobile devices, smart phones, wearable devices, etc.) operated by individuals within the channels **170**. For example, these devices (or applications installed thereon) can generate channel events **175** indicating locations of the devices, transactions conducted using the devices, and/or other information.

[0059] Additional channel events **175** can be received one or more external data sources **130**, which can include third-party websites, databases, and/or servers that provide information relating to the channels **170** and/or individuals located within the channels **170**. Exemplary external data sources **130** can include websites, databases, and/or servers associated with cellular device providers, weather outlets, news outlets, social media sites, and/or the like. In some embodiments, these and other external data sources **130** be used to derive or generate channel events **175** relating to weather conditions within the channels **170**, events occurring with the channel, locations of individuals within the channels, etc.

[0060] FIG. 5A is block diagram illustrating examples of channel events **175** that can be received by each of the nodes **165**. As shown, the channel events **175** can include, inter alia, location data **501**, transaction data **502**, user demographic data **503**, merchant data **504**, weather data **505**, and event data **506**.

[0061] The location data **501** received by each node **165** can indicate the current locations and/or historical locations and movements of individuals (or computing devices **110** operated by individuals) located in a channel **170** associated with the node **165**. For example, the location data **501** may include GPS coordinates indicating the current locations of individuals, and previous locations of those individuals. In some embodiments, the location data **501** can be received directly from computing devices **110** operated by the individuals and/or an external data source **130**, such as a cellular service provider.

[0062] The transaction data **502** received by each node **165** can indicate purchases that are made within a channel

170 associated with the node **165** and/or purchases made by individuals that are currently located within the channel **170**. The transaction data **502** also may indicate transaction patterns or profiles for each of the individuals located in the channel **170** (e.g., indicating the types of products or services routinely purchased by the individual and/or the types of businesses routinely frequented by the individual). In some cases, the transaction **502** data also may indicate the channel **170** where each transaction was conducted. In some embodiments, the transaction data **502** can be received directly from computing devices **110** operated by the individuals and/or an external data source **130**, such as a third-party merchant system, credit card service provider, digital payments provider, etc.

[0063] The demographic information **503** received by each node **165** can include various characteristics or attributes relating to individuals within a corresponding channel **170** associated with the node **165**. For example, the user demographic data **503** for each individual can indicate some or all of the following: age, sex, race, ethnicity, income, marital status, occupation, education level, interests, etc. In some embodiments, the demographic information **503** can be received directly from computing devices **110** operated by the individuals and/or an external data source **130**, such as a social media sites, marketing or advertising services, etc.

[0064] The merchant data **504** received by each node **165** can provide information related to merchants (e.g., businesses, vendors, etc.) located in a channel **170** associated with the node **165**. For example, the merchant data **504** may identify the locations of the merchants, the vertical associated with the merchants, hours of operation, and products or services offered by the merchants. In some embodiments, the merchant data **504** can be received directly from computing devices **110** operated by the merchants and/or an external data source **130**, such as a crowd-sourced business review applications, business information databases, etc.

[0065] The weather data **505** received by each node **165** can indicate the current weather conditions and/or historical weather conditions in a channel **170** associated with the node **165**. The weather data **505** also may indicate forecasts of future weather conditions for the channel **170**. In some embodiments, the weather data **505** can be received from one or more external data sources **130**, such as those that provide weather forecasting services.

[0066] The event data **505** received by each node **165** can provide information associated with events (e.g., concerts, conventions, seminars, shows, etc.) in the channel **170**. The event data **505** can include information identifying ongoing events, as well as previously held or upcoming events. The event data **505** may include dates and times associated with each of the events **505**. In some embodiments, the event data **505** can be received directly from computing devices **110** operated by event providers and/or an external data source **130**, such as a social media sites, community bulletin board sites, etc.

[0067] The categories identified in FIG. 5A are intended to provide examples of content that may be included in channel events **175** received by each node **165** of the decentralized computing network **160**. However, it should be recognized that the channel events **175** received by each node **165** can include additional information or data related to the activities, individuals, entities, and/or conditions of a channel **170**.

[0068] Returning to FIGS. 1A and 1B, the channel events 175 associated with a channel 170 can be received by a node 165 dedicated or assigned to that channel 170. The node 165 can execute a channel analysis function 166 to analyze the channel events 175, and generate channel analysis data 180 for the channel 170. The channel analysis data 180 can include various types of metrics or data useful for understanding the conditions associated with the channel 170. In many scenarios, the channel analysis data 180 can include real-time information regarding the current channel conditions and/or predictions related to future channel conditions.

[0069] FIG. 5B is a block diagram illustrating examples of channel analysis data 180 that can be generated by each node 165. In some embodiments, the channel analysis function 166 executed by each node 165 may generate channel analysis data 180 that includes population surge metrics 511, movement tracking metrics 512, and demand metrics 513.

[0070] The population surge metrics 511 generated by each node 165 can detect occurrences or situations in which the number of individuals located within a channel 170 (or certain regions in the channel 170) significantly increases and/or decreases. In some embodiments, the population surge metrics 511 can include data that identifies densities of individuals throughout a channel 170. For example, the population surge metrics 511 can indicate regions within the channel 170 where there is a high density of individuals and/or regions within the channel where there are lower densities of individuals. In some cases, the population surge metrics 511 for a channel 511 can be generated, at least in part, by monitoring locations (e.g., GPS coordinates) of individuals' computing devices 110 (e.g., smart phones or mobile devices), determining the number of computing devices 110 that are located within the channel 170 (or region within the channel), and comparing that number to a value indicating an average or baseline population for the channel 170 (or region within the channel).

[0071] Each node 165 also can be configured to predict future channel conditions in which the population in the channel 170 will vary from the average or baseline population. These predictions can be generated based, at least in part, on an analysis of the channel events 175. For example, the event data 506 included in the channel events 175 can be utilized to identify times and locations in which a population surge is likely to occur in the channel 170 (or regions within the channel 170) as the result of an upcoming event (e.g., such as a concert). Additionally, or alternatively, the location data 501 included in the channel events 175 can be used to predict upcoming expansions or retractions of populations within a channel 170 (or regions within a channel 170) based on an analysis of individuals' current locations and/or based on historical movement patterns. Additionally, or alternatively, the weather data 505 included in the channel events 175 also may be utilized to predict the population surge metrics 511 based on expected weather conditions for future time periods.

[0072] The movement tracking metrics 512 generated by each node 165 can include various metrics indicating and/or predicting the locations or movements of individuals within the channel 170 associated with the node 165. The movement tracking metrics 512 can include data indicating where individuals have moved within the channel, and locations where those individuals originated. The movement tracking metrics 512 also include data that indicates movement or migration patterns of individuals (both intra-channel move-

ment patterns and inter-channel movement patterns). For example, based on an analysis of historical location data 501, the movement tracking metrics 512 may indicate historical movement patterns of individuals within the channel 170.

[0073] The movement tracking metrics 512 also can include data that predicts the movements of individuals throughout a channel 170 in a future time period. For example, the movement tracking metrics 512 can predict regions within a channel where individuals are likely like to migrate. The location tracking metrics 512 can be generated based, at least in part, on the channel events 175 (e.g., such as the location data 501 indicating current and historical locations of individuals, and event data 506 indicating current or upcoming events within the channel 170).

[0074] The demand metrics 513 generated by each node 165 can include various metrics indicating and/or predicting supply and/or demand within the channel 170 for various inventory items (e.g., products and/or services) in one more verticals (e.g., riding sharing, hotel accommodations, parking garages, restaurants, bars, retail, etc.). The current or future supply and/or demand for inventory may be determined based, at least in part, on an analysis of the population surge metrics 511 (e.g., which can indicate or predict the densities of individuals within the channel 170 or regions within the channel 170). Additionally, or alternatively, the supply and/or demand for inventory also may be based on an analysis of channel events 175 received by the node 165, such as channel events that include transaction data 502 (e.g., which may indicate recent or historical purchases made within the channel 170), weather data 505 (e.g., which may indicate weather conditions affecting the demand for inventory), merchant data 504 (e.g., which may indicate the supply of inventory in the channel 170), and/or other channel events 175.

[0075] The population surge metrics 511, movement tracking metrics 512, and demand metrics 513 are provided as examples of channel analysis data 180 that can be generated by a node 165. However, it should be recognized that the channel analysis data 180 can include many other types of metrics or analytics relevant to the conditions of the channels 170.

[0076] Returning to FIGS. 1A and 1B, the one or more computing devices 110 can enable individuals to access the analytics platform 150 over the network 105 (e.g., over the Internet via a web browser application). For example, after a user account is established with the analytics platform 150, a user may utilize the analytics platform 150 to access to channel analysis data 180 generated by the analytics platform 180. In some embodiments, a user may be provided with access to the channel analysis data 180 generated across all of the channels 170. In other embodiments, a user may designate particular channels 170 of interest and receive channel analysis data 180 from the designated channels 170. In some embodiments, a user also may designate particular verticals or industries of interest, and receive channel analysis data 180 pertaining specifically to those verticals or industries.

[0077] The analytics platform 150 may generate various graphical user interfaces (GUIs) that display the channel analysis data 180 and/or other associated data (e.g., channel event information, user account profiles, etc.), and these interfaces can be accessed via the user accounts. The interfaces provided by the analytics platform 150 also can

include selectable options for configuring one or more deployment functions **190**. The deployment functions **190** can permit users to leverage the channel analysis data **180** (and other data generated by the analytics platform **150**) for various purposes.

[0078] One exemplary deployment function **190** can include a notification function **191**. The notification function **191** enables users to configure the transmission of notifications in various scenarios. The notifications can be transmitted in various ways (e.g., via e-mail, cellular text messages, inbox messages on user accounts, data presented on GUIs, etc.) and the notifications can be sent to various devices (e.g., client systems **140**, mobile or computing devices operated by individuals located in a channel **170**, etc.).

[0079] In some cases, the notification function **191** can be configured to periodically transmit notifications that include channel analysis data **180**, including both real-time and predictive channel analysis data **180**. In this scenario, the notifications may provide daily, weekly, monthly and/or annual summaries of channel analysis data **160** relevant to specific users. The notification function **191** also can be configured to send notifications in response to trigger events that are defined or configured by users. In this scenario, users may specify or define triggering criteria, and the analytics platform **150** may automatically send notifications to computing devices **110** and/or client systems **140** when specific trigger events are detected or predicted by the analytics platform **150** (e.g., in response to detecting or predicting a surge in population within the channel and/or a surge in demand for inventory). The notification function **191** also can be configured to send notifications to individuals (e.g., their mobile devices) located within a channel **170**. In this scenario, the notifications can include information about merchants located in a channel **170** and, in some cases, can include offers and discounts for various products and services.

[0080] Another exemplary deployment function **190** can include an interfacing function **192**. The interfacing function **192** can permit a user to interface the analytics platform **150** with various external applications and/or systems, thereby enabling those applications and/or systems to receive and utilize the channel analysis data **180** generated by the analytics platform **150**.

[0081] In some embodiments, one or more of applications running on, or operated by, the client systems **140** can be directly interfaced with the analytics platform **150** (e.g., via an application programming interface or API provided by the analytics platform **150**). In some exemplary scenarios, an client system **140** may execute or provide a ride hailing application, a lodging booking application (e.g., hotel booking application), a dining reservation application (e.g., an application for scheduling dining reservations), a ticket booking application (e.g., for purchasing tickets to concerts, sporting games, and/or other events), a pricing application (e.g., an program that computes or determines prices for products and/or services), a staffing application (e.g., a program that schedules employees), and/or an inventory management application (e.g., a program that allocates inventory among different channels **170** or locations, places orders for new inventory, etc.). The interfacing function **192** can connect the analytics system **150** to these applications (and other types of applications), thereby enabling the applications to directly receive the channel analysis data **180**

and utilize the channel analysis data **180** to automate control of one or more functions (e.g., such as determining or adjusting pricing information, adjusting inventory allocations, initiating purchases of additional inventory, adjusting staffing at locations, etc.).

[0082] In one example, the deployment functions **190** can be utilized to implement surge pricing functions, which change or adjust the prices of products and/or services based on a supply or demand for those products and/or services. For example, the population surge metrics **511** and/or demand metrics **511** included in the channel analysis data **180** for a channel **170** can be used to dynamically adjust the pricing for hotel rooms, ride hailing services, parking garage spaces, and/or other types of inventory based on the demand for the inventory within the channel **170**. Similarly, the movement tracking metrics **512** can be utilized to detect particular regions within a channel **170** where demand is likely to be higher and to adjust prices accordingly. These surge pricing functions can be automated by interfacing client systems **140** with the analytics platform **150** (e.g., via the interfacing function **192**) and/or they can be performed manually based on a review of the channel analysis information **180**.

[0083] In another example, the deployment functions **190** can be utilized by client systems **140** to reallocate inventory or resources based on the channel analysis data **180**. For example, a client system **140** that provides ride-sharing services can reallocate drivers to regions or areas where demand is higher and/or expected to be higher. Along similar lines, a client system **140** that is affiliated with a restaurant within a channel **170** can place orders for additional inventory and/or adjusting staffing in scenarios where demand is higher and/or expected to be higher. The channel analysis data and deployment functions **190** can be utilized in many other ways to enhance operations of merchants within the channels **170**.

[0084] FIG. 2A illustrates an exemplary decentralized computing network **160** comprising a plurality of nodes **165** arranged in a geographical hierarchy **210**. In some embodiments, the geographical hierarchy **210** comprises a plurality of layers (such as exemplary layers A, B, C, D . . . N), and each of the layers can include one or more nodes **165**. In general, the nodes **165** located in the upper layers of the geographical hierarchy **210** encompass broader channels **170** or geographic areas than those included in lower layers.

[0085] To illustrate by way of example, Layer A may include a global node **165** that corresponds to a global channel (e.g., channel **170A**), which encompasses all geographic regions that are analyzed by the analytics platform **150**. Layer B can include a plurality of nodes **165** that correspond to country-level channels (e.g., such as channel **170B**), each of which encompasses a geographic region associated with a particular country. Layer C can include a plurality of nodes **165** that correspond to state-level channels (e.g., such as channel **170C**), each of which encompasses a geographic region associated with a particular state, province, district or similar type of region within a country. Layer D can include a plurality of nodes that correspond to city-level channels (e.g., such as channel **170D**), each of which encompasses a geographic region associated with a particular city, town, village, or the like. Layer E can include a plurality of nodes that correspond to even more granular channels (e.g., such as channel **170E**), each of which encom-

passes a geographic region (e.g., a neighborhood or area) within a particular city, town, village, or the like.

[0086] Parent-child relationships **220** can be established among related nodes **365** in the geographical hierarchy **210**. A child node can represent a channel **170** (or geographic region) that is a subset of a channel **170** associated with a parent node. Conversely, a parent node can represent a channel **170** (or geographic region) that is a superset of one or more channels **170** associated with child nodes. Thus, each node may be connected to parent node in an upper layer if that node is a subset of the node in the upper layer, and each node can be connected to one or more child nodes in a lower layer if that node is a superset of the nodes in the lower layer.

[0087] For example, a first node in Layer C can be associated with a channel covering the state of New York. Therefore, this node can be connected to a parent node in Layer B that corresponds to a channel covering the United States, and can be connected to a plurality of child nodes in Layer D corresponding to cities in the state of New York (e.g., New York City, Buffalo, Rochester, etc.). Similarly, a second node in Layer C may correspond to a channel covering the province of British Columbia. This node can be connected to a parent node in Layer B that corresponds to a channel encompassing the country of Canada, and can be connected a plurality of child nodes in Layer D corresponding to cities in the province of British Columbia. Along similar lines, the nodes in Layer E can correspond to specific regions (e.g., blocks, neighborhoods, boroughs, etc.) within particular cities, and can be connected to corresponding nodes in Layer D.

[0088] A node communication protocol **167** (FIG. 1B) can enable communication among the nodes **165** in the decentralized computing network **160** based on the parent-child relationships **220** among the nodes. For example, a given node can exchange information with one or more parent nodes and/or one or more child nodes based on geographical associations or relationships. The exchange of information among the nodes can be used, at least in part, to generate the channel analysis data **180** across the nodes **165** in a manner that avoids redundancies and reduces storage and processing requirements.

[0089] To illustrate by way of example, consider a scenario in which a node **165** in Layer C (e.g., corresponding to a channel **170** for the State of New York) is computing demand metrics **513** that indicate or predict the demand for one or more products or services in the channel **170**. In this scenario, the node communication protocol **167** enables the node in Layer C to communicate with and receive demand metrics **513** generated by the nodes in Layer D (e.g., that can correspond to channels for cities within the State of New York). The node in Layer C can compute the demand metrics **513** for its channel **170**, at least in part, using the demand metrics received from the nodes in Layer D. Along similar lines, the nodes in Layer D can compute demand metrics **513** for corresponding channels **170** using the demand metrics **513** that were computed by corresponding child nodes in Layer E.

[0090] In this manner, the nodes **165** are able to leverage the data generated by the related nodes and avoid processing the data associated with those nodes, thereby avoiding redundancies in computations. The node communication protocol **167** can enable the nodes to leverage the parent-child relationships **220** to compute other metrics included

with the channel analysis data **180** (e.g., such as population surge metrics **511**, movement tracking metrics **512**, and/or other metrics described herein).

[0091] Additionally, in some embodiments, the node communication protocol **167** can include or utilize a blockchain protocol to record and track communications among the nodes **165** of the decentralized computing network **160**. For example, each of the nodes **165** can store and maintain a distributed ledger or database that records the data exchanged among the nodes **165** (e.g., which may include channel analysis information **180** and/or channel events **175**). The distributed ledger or database can be updated with a new block or entry each time data from one node **165** is passed to another node **165**, and the block or entry can be appended with various attributes (e.g., data indicating the node receiving information, the node sending information, the information that was exchanged, the date/time of exchange, etc.). In some scenarios, a consensus mechanism or protocol may be utilized to approve the addition of new blocks to the distributed ledger or database and/or to ensure a single, valid copy of the distributed ledger or database is shared by all of the nodes **165**.

[0092] For ease of understanding, the examples discussed above (and in other portions of this disclosure) describe geographical hierarchies **210** or channels **170** that are based on political boundaries, such as those corresponding to countries, states, cities, etc. However, it should be recognized that the geographical hierarchies **210** and channels **170** can be defined in many other ways that are not associated with political boundaries. For example, the channels **170** associated with the nodes **165** can correspond to arbitrarily selected geographic regions that are not tied to political boundaries and/or can correspond to geographic regions that are selected based on longitude and latitude coordinates.

[0093] FIG. 2B illustrates an exemplary cloud-based computing environment **200** that is configured to host and/or implement the decentralized computing network **160**. In certain embodiments, the cloud-based computing environment **200** can include a plurality of servers (e.g., such as servers **120**) that provide, inter alia, a shared pool of processing, memory, and/or storage resources.

[0094] In this exemplary environment, each of the nodes **165** included in the decentralized computing network **160** can be executed on a separate virtual machine (VM) **230** (e.g., such as virtual machines **1**, **2**, . . . **N**). Each of the virtual machines **230** can utilize dedicated memory resources **201** and dedicated processing resources **202** to execute the functionalities associated with a corresponding node **165**. The dedicated memory resources **201** can include one or more computer storage devices (e.g., such as computer storage devices **101**) selected from a shared pool of server resources. The dedicated processing resources **202** can include one or more processing devices (e.g., such as processing devices **102**) selected from a shared pool of server resources.

[0095] In certain embodiments, each of the virtual machines **230** can represent a process virtual machine (also referred to as an application virtual machine) that executes the functions associated with a node **165**, including receiving channel events **175**, executing the channel analysis function **166**, and generating channel analysis information **180**. As explained above, a node communication protocol **167** enables the nodes **165** (and corresponding virtual

machines 230) to communicate with each other for generating channel analysis data 180.

[0096] The node manager 250 can operate as a virtual machine manager (VMM) and/or hypervisor that manages the creation and execution of the virtual machines 230 in the cloud-based computing environment 200. Additionally, or alternatively, the node manager 250 can perform functions associated with allocating or assigning the nodes 165 to the virtual machines 230.

[0097] The node manager 250 also can perform operations associated with allocating resources (e.g., dedicated memory resources 201, dedicated processing resources 202, storage resources, etc.), to each of the virtual machines 230 and/or to each of the nodes 165 associated with the virtual machines 230. In some instances, the node manager 250 can monitor the workloads associated with each of the nodes 165 and dynamically adjust or allocate memory resources 201 and processing resources 202 to the nodes 165 (or virtual machines 230 executing the nodes 165) based on the workloads of the nodes 165. For example, in scenarios in which a node 165 has a large workload (e.g., because it is required to process data for large populations of individuals), the node manager 250 can allocate additional memory resources 201 and processing resources 202 to the node 165.

[0098] In certain embodiments, the node manager 250 also can perform operations that allocate nodes 165 to the virtual machines 230 in the cloud-based computing environment 200. In some instances, the node manager 250 may allocate one node to each of the virtual machines 230. In other instances, the node manager 250 may allocate a plurality of nodes to some or all of the virtual machines 230.

[0099] Additionally, the node manager 250 can perform operations that associate nodes 165 with the channels 170. In some embodiments, the node manager 250 can associate one node 170 with each of the channels 170 and, in other instances, the node manager 250 can allocate a plurality of nodes 165 to each of the channels 170 or some of the channels 170. For example, in scenarios where the workload of node 165 is high, the node manager 250 can allocate additional nodes 165 to help process the workload for a corresponding channel 170.

[0100] The decentralized computing network 160 can be implemented in other environments and configurations. For example, the decentralized computing network 160 can be implemented in alternative cloud-based computing environments and configurations. Additionally, the decentralized computing network 160 can be implemented without the usage of any cloud-based computing environment. For example, in some embodiments, a plurality of hardware servers, each of which is dedicated to execute the functionality of a corresponding node 165, can be utilized to implement the decentralized computing network 160. The decentralized computing network 160 can be implemented in other ways as well.

[0101] FIG. 3 is an illustration demonstrating channel analysis data 180 being generated across various multiple verticals 310 according to certain embodiments. The decentralized computing network 160 includes a plurality of nodes 165, each of which are dedicated to channel 170. Each node 165 receives various channel events 175 corresponding to its dedicated channel 170. Each node 165 can process and analyze the channel events 175 to generate channel analysis data 180, which can include real-time metrics pertaining to

current channel conditions and/or predictive metrics pertaining to future channel conditions.

[0102] This real-time data and/or predictive data can be generated for a plurality of verticals 310 (e.g., business sectors or industries), including those related to transportation, lodging, restaurants, ticket booking, parking services, etc. In certain embodiments, the channel analysis data 180 can be customized specifically to each of the verticals 310. In some scenarios, the channel analysis data 180 can separately compute demand metrics 513 for each vertical 310 and/or for inventory items associated with each vertical 310. For example, demand metrics 513 can be separately computed for inventory items corresponding to room availabilities for a hotel or lodging vertical, vehicle availabilities for a ride hailing or transportation vertical, parking space availabilities for a garage or parking vertical, reservation availabilities for a restaurant or dining vertical, etc.

[0103] As mentioned above, the analytics platform 150 can generate notifications and/or GUIs that display or include some or all of the channel analysis data 180. In some cases, these GUIs and notifications can be customized or tailored for to present metrics or data for specific verticals 310. Similarly, any data provided directly to client systems 140 with the analytics platform 150 can be customized or tailored for specific verticals 310 and/or specific client systems 140.

[0104] FIG. 4 is an illustration demonstrating aspects of a real-time engine 410 and a predictive engine 420 according to certain embodiments. In certain embodiments, the real-time engine 410 and the predictive engine 420 can represent subroutines that are executed by a channel analysis function 166 for a node 165 to generate channel analysis data 180 for a channel 170 associated with the node 165.

[0105] The real-time engine 410 can be configured to generate channel analysis data 180 pertaining to the current or real-time conditions of the channel 170, and the predictive engine 420 can be configured to generate channel analysis data 180 pertaining to predicted conditions of the channel 170 in one or more future time periods. For example, the real-time engine 410 can generate population surge metrics 511, movement tracking metrics 512, and demand metrics 513 indicating the real-time conditions of the channel 170, while the predictive engine 420 can generate population surge metrics 511, movement tracking metrics 512, and demand metrics 513 predicting conditions of the channel 170 in one or more future time periods. As explained above, the real-time engine 410 and predictive engine 420 can utilize various channel events 175 (e.g., indicating spending habits, schedule events, weather conditions, historical movements of individuals, population density, etc.) to determine the current and future conditions of the channels.

[0106] FIG. 6 is a block diagram illustrating exemplary features, components, and/or functions of a client system 140 according to certain embodiments. The client system 140 can include one or computing devices 110 and/or one or more servers 120, each of which includes one or more computer storage devices 101 and one or more processor devices 102. The client system 140 can host and execute one or more client applications 610.

[0107] Exemplary client applications 610 provided by a client system 140 can include one or more of the following: 1) a ride hailing application 611 (e.g., an application that connects passengers with drivers to schedule rides); 2) an accommodation application 612 (e.g., an application that

permits guests to schedule rooms or lodging); 3) a travel application **613** (e.g., an application that permits individuals to book schedule transportation with airlines, trains, buses, boats, etc.); and 4) a reservation application **614** (e.g., an application that permits individuals to schedule reservations or tickets for restaurants, concerts, events, bars, parking spaces, and/or other venues). Other types of client applications **610** also may be hosted and executed by the client systems **140**. In certain embodiments, the client applications **610** can represent web-based applications that are accessible via a web browser and/or local applications (e.g., mobile apps) that is installed on devices (e.g., mobile devices or smart phones) operated by end-users.

[0108] The channel analysis data **180** (e.g., the population surge metrics **511**, movement tracking metrics **512**, demand metrics **513**, and/or other data described herein) can be utilized to enhance various functionalities of the client applications **610**. In some scenarios, the channel analysis data **180** can be utilized to enhance or implement one or more demand adjustment functions **650**, such as a pricing function **620** and/or an inventory management function **630**, for each of the client applications **160**. The demand adjustment functions **650** can be configured to adjust pricing and/or inventory information for inventory items based on actual or predicted supply metrics and/or demand metrics for the inventory items. The pricing function **620** and/or inventory management function **630** can be included with the functionality of each of the client applications **610**, or can be included in separate applications that communicate with the client applications **610**.

[0109] The pricing function **620** can utilize the channel analysis data **180** to determine pricing for one or more inventory items **635**, which may generally include any type of product or service made available by a client application. For example, depending on the functionality of a given client application **160**, the pricing function **620** can determine pricing for ride hailing services, taxi services, lodging accommodations, event tickets (e.g., for sporting events or concerts), airline tickets, train tickets, packing spaces, etc.

[0110] In some scenarios, the channel analysis data **180** can be utilized to implement a surge pricing function **625** for one or more of the client applications **610**. A surge pricing function **625** generally represents a function that adjusts the price of one or more inventory items **635** based on the demand for the inventory items **635** (e.g., based on a comparison of the supply and the demand for the inventory items **635**). The demand metrics **513** (and/or other channel analysis data **180**) generated by the analytics platform **150** may be utilized to dynamically adjust the pricing of one or more inventory items **635** offered by each of the client applications **610**.

[0111] The inventory management function **630** can utilize the channel analysis data **180** to manage or adjust inventory items **635** in various ways. For example, the inventory management function **630** can detect when additional inventory items **635** should be ordered to accommodate a current demand for inventory items **635** and/or a predicted future demand for inventory **635**. In some scenarios, the inventory management function **630** also can be configured to automatically place an order for additional inventory items **635** to accommodate a spike in a current or predicted demand for the inventory items **635**.

[0112] The inventory management function **630** also can reallocate inventory items **635** to accommodate varying

supply and demand metrics across different channels **170** and/or within a given channel **170**. For example, in some scenarios, a merchant may have multiple business locations, including multiple locations within a given channel **170** and multiple locations situated outside the channel **170**. The inventory management function **630** can utilize the channel analysis data **180** to dynamically reallocate inventory items **635** among the business locations in within the given channel **170** to accommodate the varying demands at those locations and/or to maximize sales of inventory items **635** across all locations. Similarly, the inventory management function **630** can utilize the channel analysis data **180** to dynamically reallocate inventory items **635** from a location in one channel **180** to one or more separate channels in order to accommodate the varying demands in each channel **170** and/or to maximize sales of inventory items **635** across all channels **170**.

[0113] In certain embodiments, the client applications **610** can additionally, or alternatively, be stored on and executed by the analytics platform **150**. Similarly, the demand adjustment functions **650** (e.g., including the pricing function **620**, the surge pricing function **625**, and/or the inventory management function **630**) can be stored on and executed by the analytics platform **150**.

[0114] For example, in certain embodiments, the decentralized computing network **160** can be utilized to execute the demand adjustment functions **650**. In addition to generating the channel analysis data **180**, each node **165** of the decentralized computing network **160** can execute one or more demand adjustment functions **650** for a channel **170** corresponding to the node **165**. In one example, each node **165** can execute a surge pricing function **625** that utilizes the channel analysis data (e.g., the population surge metrics **511** and/or demand metrics **513**) to dynamically adjust the pricing of one or more inventory items **635**. In another example, each node **165** can execute an inventory management function **630** that dynamically allocates or reallocates inventory items **635** within a channel **170** and/or among a plurality of channels **170**.

[0115] FIGS. 2C and 2D illustrate examples of a decentralized computing network **160** that is configured to execute demand adjustment functions **650**.

[0116] In FIG. 2C, four nodes **165** are identified by cross-hashing to indicate that the nodes **165** have detected channel conditions indicating the occurrence of a demand surge **290** for one or more inventory items **635** (or a demand surge **290** in one or more verticals **310**) in their corresponding channels **170** (channels **170C**, **170F**, **170G**, and **170H**). The demand surge **290** can indicate an upward fluctuation in the demand for the one or more inventory items **635** (e.g., in scenarios where there is a high demand or demand is increasing) and/or a downward fluctuation in the demand for the one or more inventory items **635** (e.g., in scenarios where there is low demand or demand is decreasing). The other nodes **165** (without cross-hashing) correspond to nodes that have not detected a demand surge **290** in their corresponding channels **170**.

[0117] In some cases, each node **165** may detect the occurrence of demand surge **290** by comparing a first value indicating an actual or predicted demand for one or more inventory items **635** (which may be included in the demand metrics **513** computed by the node **165**) with a second value indicating a baseline or standard demand for a channel **170**. Additionally, or alternatively, each node **165** can detect the

occurrence of demand surge 290 by comparing the first value with a threshold value (e.g., such that a demand surge 290 is detected when the first value exceeds or is below the threshold value). The demand surge 290 can be detected in other ways as well.

[0118] In FIG. 2D, each of the nodes 165 included in the decentralized computing network 160 have computed a multiplier (or other value) to be utilized in connection with one or more demand adjustment functions 650. Nodes 165 that have not detected a demand surge 290 indicating an upwardly trending demand in corresponding channels 170 are shown with a multiplier of one ($\times 1$). The four nodes identified by cross-hatching in FIG. 2C (i.e., the nodes 165 that have detected an upwardly trending demand surge 290 in their corresponding channels 170) have computed greater multipliers ($\times 1.4$, $\times 2.0$, $\times 2.4$, and 1.2), each of which corresponds to the severity or level of the demand surge 290 in a corresponding channel 170.

[0119] The multipliers computed by the nodes can be utilized by dynamically modify or update one or more demand adjustment functions 650. For example, consider a scenario in which the nodes 165 are executing a surge pricing function 625 that dynamically adjusts prices for one or more inventory items 635 in channels 170 corresponding to the nodes 165. In this scenario, the multipliers can be utilized to increase the prices for the one or more inventory items 635. Each node 165 may determine an adjusted price for the one or more inventory items 635 by multiplying a baseline price with the multiplier that was computed by the node 165. In some instances, the adjusted price, which reflects an adjustment based on the demand surge, can be utilized by a client application 160 to price the one or more inventory items 635 (e.g., to adjust prices for ride hailing services, hotel rooms, event tickets, etc.).

[0120] The multipliers computed by the nodes 165 also can be utilized by an inventory management function 630 to dynamically adjust allocations of inventory items 635. For example, consider a scenario in which a ride hailing service is experiencing a population surge or demand surge in a first set of channels 170, while population or demand conditions in a second set of channels 170 remains normal or constant. In this scenario, the ride hailing service may wish to reallocate inventory items 635 (e.g., drivers) from the channels 170 with lower demand to the channels 170 that are experiencing greater demand. The multipliers computed by the nodes 165 can be utilized to determine the amount of inventory that should be transferred to the channels 170 with greater demand.

[0121] In some embodiments, each of the nodes 165 also can be configured to detect or determine a supply surge (rather than a demand surge 290, or in addition to calculating the demand surge 290). The supply surge can indicate an upward fluctuation in the supply for the one or more inventory items 635 (e.g., in scenarios where there is a high supply or the supply is increasing) and/or a downward fluctuation in the supply for the one or more inventory items 635 (e.g., in scenarios where there is low supply or supply is decreasing). This supply surge can be utilized in a similar manner to the demand surge 290, such as to implement pricing adjustment functions and/or inventory management functions. For example, a high supply surge may result in lowering prices or inventory levels, while a low supply may result in raising prices or inventory levels.

[0122] In some instances, the supply surge or supply-related information can be utilized in computing the demand surge 290. For example, the detection of a demand surge 290 can involve consideration of both the supply and demand for one or more inventory items 635, and the one or more demand adjustment functions 610 can adjust pricing information and/or inventory allocations based on a joint consideration of the supply and demand information for the one or more inventory items 635.

[0123] FIG. 7 illustrates a flow chart for an exemplary method 700 according to certain embodiments. Method 700 is merely exemplary and is not limited to the embodiments presented herein. Method 700 can be employed in many different embodiments or examples not specifically depicted or described herein. In some embodiments, the steps of method 700 can be performed in the order presented. In other embodiments, the steps of method 700 can be performed in any suitable order. In still other embodiments, one or more of the steps of method 700 can be combined or skipped. In many embodiments, system 100, analytics platform 150, and/or decentralized computing network 160 can be configured to perform method 700 and/or one or more of the steps of method 700. In these or other embodiments, one or more of the steps of method 700 can be implemented as one or more computer instructions configured to run at one or more processing devices 102 and configured to be stored at one or more non-transitory computer storage devices 101. Such non-transitory memory storage devices 101 can be part of a computer system such as system 100, analytics platform 150, and/or decentralized computing network 160.

[0124] In step 710, a decentralized computing architecture or network 160 comprising a plurality of nodes 165 is provided. In certain embodiments, the nodes 165 included in the decentralized computing architecture 160 can be arranged in a geographical hierarchy 210 that links the nodes 165 using parent-child relationships 220 based on geographical associations.

[0125] In step 720, each node 165 of the decentralized computing architecture or network 160 is associated with a separate channel 170 corresponding to a geographic region. For example, each node 175 in the decentralized computing architecture 160 can be dedicated to a particular channel 170 such that the node 175 receives and processes data exclusively and/or independently for the channel 170.

[0126] In step 730, each node 165 of the decentralized computing architecture or network 160 receives and analyzes channel events 175 for the channel 170 associated with node 165. For example, each node 165 can receive various types of channels events 175 relating to conditions of an associated channel 170. Exemplary channel events 175 can include location data 501, transaction data 502, user demographic data 503, merchant data 504, weather data 505, and/or event data 505.

[0127] In step 740, each node 165 generates channel analysis data 180 corresponding to the channel 170 associated with the node 165 based, at least in part, on an analysis of the channel events 175 corresponding to the channel 170 associated with the node 165. The channel analysis data 180 can include various metrics including, but not limited to, population surge metrics 511, movement tracking metrics 512, and/or demand metrics 513. The channel analysis data 180 can include both real-time data that indicates current conditions of the channel 170 and/or predictive data that predicts future conditions of the channel.

[0128] In some embodiments, the decentralized computing network 160 can utilize the channel analysis data 180 to execute one or more demand adjustment functions, such as a pricing function 620, surge pricing function 625, and/or inventory management function 630. In certain instances, each of the nodes 165 in the decentralized computing network 160 can analyze the channel analysis data 180 (e.g., the demand metrics 513) to determine if a demand surge 290 exists for an inventory item 635 in a corresponding channel 170. In response to detecting a demand surge 290 in a channel 170, the node 170 associated with the channel 170 can determine a multiplier (or other value) that is used to determine and/or adjust the price of inventory item 635 or used to adjust allocations of the inventory items 635.

[0129] In step 750, one or more deployment functions 190 can be executed to enable the channel analysis data 180 to be accessed and/or utilized by one or more client systems 140. The deployment functions 190 can enable the client systems 140 to utilize the channel analysis data 180 to enhance the functionalities of one or more client applications 610. One exemplary deployment function 190 can include a notification function 191 that transmits notifications to the one or more client systems 140 and/or displays the notifications on one or more GUIs that are accessible via an analytics platform 150. Another exemplary deployment 190 can include an interfacing function 192 that interfaces the analytics platform 150 with various the client applications 610 and/or client systems 140, thereby enabling those external applications and/or systems to directly access the channel analysis data 180. In some instances, a client application 610 that is interfaced with the analytics platform 150 can utilize the channel analysis data to execute a pricing function 620, surge price function 625, and/or an inventory management function 630. As evidenced by the disclosure herein, the inventive techniques set forth in this disclosure are rooted in computer technologies that overcome existing problems in known networking and data processing systems, including problems dealing with utilizing a centralized network architecture to process data over multiple channels or geographic regions. The techniques described in this disclosure provide a technical solution (e.g., that utilizes improved decentralized computing networks) for overcoming the limitations associated with known techniques. This technology-based solution marks an improvement over existing capabilities and functionalities related to processing data across multiple channels or geographic areas.

[0130] In certain embodiments, the techniques described herein can be utilized continuously at a scale that cannot be reasonably performed using manual techniques or the human mind. For example, in many embodiments, real-time information from large numbers of channels or geographic areas can be simultaneously processed and analyzed to provide real-time updates to client systems. This simultaneous processing of real-time data cannot be performed by a human mind.

[0131] Additionally, in certain embodiments, the techniques described herein solve a technical problem that arises only within the realm of computer networks, as decentralized computer networks or architectures do not exist outside the realm of computer networks.

[0132] Embodiments may include a computer program product accessible from a computer-usable or computer-readable medium providing program code for use by or in connection with a computer or any instruction execution

system. A computer-usable or computer-readable medium may include any apparatus that stores, communicates, propagates, or transports the program for use by or in connection with the instruction execution system, apparatus, or device. The medium can be a magnetic, optical, electronic, electromagnetic, infrared, or semiconductor system (or apparatus or device) or a propagation medium. The medium may include a computer-readable storage medium, such as a semiconductor or solid-state memory, magnetic tape, a removable computer diskette, a random access memory (RAM), a read-only memory (ROM), a rigid magnetic disk and an optical disk, etc.

[0133] A data processing system suitable for storing and/or executing program code may include at least one processor coupled directly or indirectly to memory elements through a system bus. The memory elements can include local memory employed during actual execution of the program code, bulk storage, and cache memories that provide temporary storage of at least some program code to reduce the number of times code is retrieved from bulk storage during execution. Input/output or I/O devices (including but not limited to keyboards, displays, pointing devices, etc.) may be coupled to the system either directly or through intervening I/O controllers.

[0134] Network adapters may also be coupled to the system to enable the data processing system to become coupled to other data processing systems or remote printers or storage devices through intervening private or public networks. Modems, cable modems, and Ethernet cards are just a few of the currently available types of network adapters.

[0135] It should be recognized that any features and/or functionalities described for an embodiment in this application can be incorporated into any other embodiment mentioned in this disclosure. Moreover, the embodiments described in this disclosure can be combined in various ways. Additionally, while the description herein may describe certain embodiments, features, or components as being implemented in software or hardware, it should be recognized that any embodiment, feature, or component that is described in the present application may be implemented in hardware, software, or a combination of the two.

[0136] While various novel features of the invention have been shown, described, and pointed out as applied to particular embodiments thereof, it should be understood that various omissions and substitutions, and changes in the form and details of the systems and methods described and illustrated, may be made by those skilled in the art without departing from the spirit of the invention. Amongst other things, the steps in the methods may be carried out in different orders in many cases where such may be appropriate. Those skilled in the art will recognize, based on the above disclosure and an understanding of the teachings of the invention, that the particular hardware and devices that are part of the system described herein, and the general functionality provided by and incorporated therein, may vary in different embodiments of the invention. Accordingly, the description of system components are for illustrative purposes to facilitate a full and complete understanding and appreciation of the various aspects and functionality of particular embodiments of the invention as realized in system and method embodiments thereof. Those skilled in the art will appreciate that the invention can be practiced in other than the described embodiments, which are presented

for purposes of illustration and not limitation. Variations, modifications, and other implementations of what is described herein may occur to those of ordinary skill in the art without departing from the spirit and scope of the present invention and its claims.

1. A computerized method for processing data across multiple channels, comprising:

providing access to an analytics platform that comprises:

(a) a decentralized computing network comprising a plurality of nodes, wherein each node of the decentralized computing network is associated with a channel corresponding to a geographic region and the plurality of nodes included in the decentralized computing network are arranged in a geographical hierarchy that defines parent-child relationships based on geographic associations; and

(b) an application programming interface (API) that facilitates a connection between the analytics platform and one or more client systems over a network;

configuring each node of the decentralized computing network to receive and analyze channel events for a channel associated with the node;

generating, by each node of the decentralized computing network, channel analysis data for the channel associated with the node, wherein the channel analysis data for the channel is generated based, at least in part, on an analysis of the channel events associated with the channel and the channel analysis data at least indicates a demand for one or more inventory items within the channel; and

transmitting, via the API of the analytics platform, the channel analysis data corresponding to the channel to the one or more client systems connected to the analytics platform, wherein the one or more client systems are configured to execute a demand adjustment function that adjusts a price of the one or more inventory items within the channel based, at least in part, on the demand for the one or more inventory items in the channel.

2. The computerized method of claim 1, wherein the method further comprises:

detecting, by at least one of the plurality of nodes, a demand surge in one or more channels based on the channel analysis data; and

transmitting, via the API of the analytics platform, data identifying the demand surge in the one or more channels to the one or more client systems connected to the analytics platform.

3. The computerized method of claim 1, wherein the decentralized computing network implements a blockchain protocol to record communications among the plurality of nodes included within the decentralized computing network.

4. The computerized method of claim 3, wherein, in accordance with the blockchain protocol, each of the plurality of nodes is adapted to store and maintain a distributed ledger or database that records data exchanged among the plurality of nodes, including any channel analysis data that is exchanged among the plurality of nodes.

5. The computerized method of claim 4, wherein the distributed ledger or database can be updated with a new block or entry each time data from one node is passed to another node.

6. The computerized method of claim 1, wherein:

each node of the decentralized computing network is configured with a real-time engine that is configured to generate real-time channel analysis data indicating current or real-time conditions of the channel corresponding to the node; and

the real-time channel analysis data corresponding to the channel is transmitted to the one or more client systems connected to the analytics platform via the API of the analytics platform.

7. The computerized method of claim 6, wherein the real-time channel analysis data at least comprises one or more real-time demand metrics indicating a current or real-time demand for the one or more inventory items within the channel.

8. The computerized method of claim 1, wherein the channel analysis data generated by the plurality of nodes included in the decentralized computing network includes demand metrics indicating or predicting the demand for inventory items included in a plurality of separate verticals within the channel.

9. The computerized method of claim 8, wherein the demand metrics indicate or predict the demand for the plurality of separate verticals including at least two of: a ride hailing vertical; a lodging or accommodation vertical; a vehicle parking vertical; a restaurant or tavern vertical; a transportation vertical; a ticketed event vertical; and an e-commerce vertical.

10. The computerized method of claim 1, wherein the channel analysis data generated by each node comprises at least two of:

population metrics predicting or indicating population fluctuations in the channel associated with the node;

movement tracking metrics predicting or indicating movements of individuals within the channel associated with the node; and

demand metrics predicting or indicating the demand for the one or more inventory items in the channel associated with the node.

11. The computerized method of claim 1, wherein each of the one or more client systems is configured to execute a surge pricing function that dynamically adjusts pricing for the one or more inventory items within the channel based, at least in part, on the channel analysis data generated for the channel.

12. A system for processing data across multiple channels, comprising:

(a) a decentralized computing network comprising one or more processing devices and one or more non-transitory computer storage devices storing computing instructions configured to be executed on the one or more processing devices and cause the one or more processing devices to process data for a plurality of nodes such that:

each node of the decentralized computing network is associated with a channel corresponding to a geographic region and the plurality of nodes included in the decentralized computing network are arranged in a geographical hierarchy that defines parent-child relationships based on geographic associations;

each node of the decentralized computing network is configured to receive and analyze channel events for the channel associated with the node; and

each node of the decentralized computing network is configured to generate channel analysis data for the

channel associated with the node, wherein the channel analysis data is generated based, at least in part, on an analysis of the channel events associated with the channel and the channel analysis data at least indicates a demand for one or more inventory items within the channel; and

- (b) an application programming interface (API) that enables a connection between an analytics platform that includes the decentralized computing network and one or more client systems over a network, wherein: the channel analysis data generated by the decentralized computing network for the channel is provided via the API to the one or more client systems connected to the analytics platform, wherein the one or more client systems are configured to execute a demand adjustment function that adjusts a price of the one or more inventory items within the channel based, at least in part, on the demand for the one or more inventory items in the channel.

13. The system of claim **12**, wherein the decentralized computing network implements a blockchain protocol to record communications among the plurality of nodes included within the decentralized computing network.

14. The system of claim **13**, wherein each of the plurality of nodes is adapted to store and maintain a distributed ledger or database that records data exchanged among the plurality of nodes, including any channel analysis data that is exchanged among the plurality of nodes, and the distributed ledger or database can be updated with a new block or entry each time data from one node is passed to another node.

15. The system of claim **12**, wherein:

each node of the decentralized computing network is configured with a real-time engine that is configured to generate real-time channel analysis data indicating current or real-time conditions of the channel corresponding to the node;

the real-time channel analysis data at least comprises real-time one or more demand metrics indicating a current or real-time demand for the one or more inventory items within the channel; and

the real-time channel analysis data corresponding to the channel is transmitted to the one or more client systems connected to the analytics platform via the API of the analytics platform.

16. The system of claim **12**, wherein the channel analysis data generated by the plurality of nodes included in the decentralized computing network includes demand metrics indicating or predicting the demand for inventory items included in a plurality of separate verticals within the channel.

17. The system of claim **16**, wherein the demand metrics indicate or predict the demand for the plurality of separate verticals including at least two of: a ride-hailing vertical; a lodging or accommodation vertical; a vehicle parking ver-

tical; a restaurant or tavern vertical; a transportation vertical; a ticketed event vertical; and an e-commerce vertical.

18. The system of claim **12**, wherein the channel analysis data generated by each node comprises at least two of:

population metrics predicting or indicating population fluctuations in the channel associated with the node;

movement tracking metrics predicting or indicating movements of individuals within the channel associated with the node; and

demand metrics predicting or indicating the demand for the one or more inventory items in the channel associated with the node.

19. The system of claim **12**, wherein each of the one or more client systems is configured to execute a surge pricing function that dynamically adjusts pricing for the one or more inventory items within the channel based, at least in part, on the channel analysis data generated for the channel.

20. A computer program product comprising one or more non-transitory storage devices that store instructions for causing one or more processing devices to:

configure a decentralized computing network with a plurality of nodes such that:

each node of the decentralized computing network is associated with a channel corresponding to a geographic region and the plurality of nodes included in the decentralized computing network are arranged in a geographical hierarchy that defines parent-child relationships based on geographic associations;

each node of the decentralized computing network is configured to receive and analyze channel events for the channel associated with the node; and

each node of the decentralized computing network is configured to generate channel analysis data for the channel associated with the node, wherein the channel analysis data is generated based, at least in part, on an analysis of the channel events associated with the channel and the channel analysis data at least indicates a demand for one or more inventory items within the channel; and

provide an application programming interface (API) that enables a connection between an analytics platform that includes the decentralized computing network and one or more client systems over a network, wherein:

the channel analysis data generated by the decentralized computing network for the channel is provided via the API to the one or more client systems connected to the analytics platform, wherein the one or more client systems are configured to execute a demand adjustment function that adjusts a price of the one or more inventory items within the channel based, at least in part, on the demand for the one or more inventory items in the channel.

* * * * *